

第九章 振动

9-1 简谐振动 振幅 周期和频率 相位.

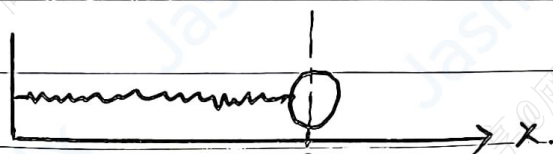
一. 简谐运动

1. 机械振动

物理量随时间的周期性变化.

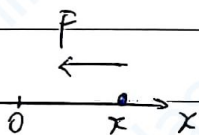
定义: 简谐运动: 物体运动时, 离开平衡位置的位移 (或位移), 可以用时间的正弦函数来描述.

二. 弹簧振子 (理想模型)



平衡位置.

$$x=0 \quad F=0$$



$$F = -kx \quad \begin{array}{ll} x > 0 & F < 0 \\ x < 0 & F > 0 \end{array}$$

方向

$$F = -kx = ma$$

$$a = -\frac{k}{m}x \quad \text{令 } \omega^2 = \frac{k}{m} \quad a = -\omega^2 x$$

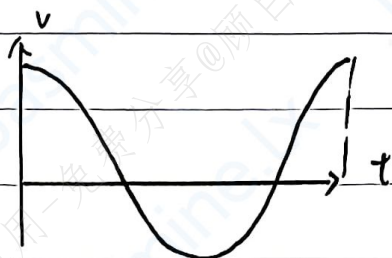
简谐振动的特征方程: $a \propto x$, 方向相反

$$a = \frac{d^2x}{dt^2} = -\omega^2 x \quad \rightarrow \quad x = A \cos(\omega t + \varphi) \quad \text{位移可用时间正弦函数描述}$$

角频率, 圆频率.

$$v = \frac{dx}{dt} = -A\omega \sin(\omega t + \varphi) = A\omega \cos(\omega t + \varphi + \frac{\pi}{2})$$

$$a = \frac{dv}{dt} = -A\omega^2 \cos(\omega t + \varphi) = A\omega^2 \cos(\omega t + \varphi + \pi)$$



三. $x = A \cos(\omega t + \varphi)$

1. 振幅 $A = |x_{\max}|$

2. 周期 频率 $T = \frac{2\pi}{\omega}$ 弹簧振子: $T = 2\pi \sqrt{\frac{m}{k}}$

$\nu = \frac{1}{T} = \frac{\omega}{2\pi}$ $\omega = 2\pi\nu = \frac{2\pi}{T}$

3. 相位 $\phi = \omega t + \varphi$ 初相位 ($t=0$) $\phi = \varphi$

① $\omega t + \varphi = 0$ $x = A$ $v = 0$ 处于正最大位移处

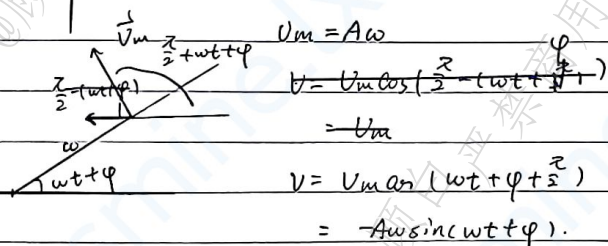
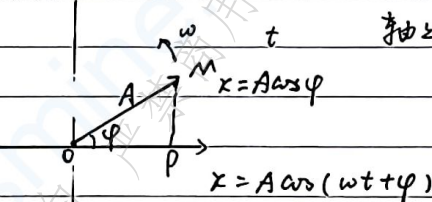
② $\omega t + \varphi = \frac{\pi}{2}$ $x = 0$ $v = -A\omega < 0$

③ $\omega t + \varphi = -\frac{\pi}{2}$ $x = 0$ $v = A\omega > 0$

$x = A \cos(\omega t + \varphi)$ $t=0$ $x = A \cos \varphi$

$v = -A\omega \sin(\omega t + \varphi)$ $v = -A\omega \sin \varphi$

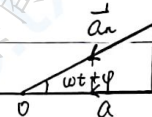
9-2 旋转矢量 P 旋转矢量 $\vec{A}$ 的端点 M 在 Ox 轴上投影点的运动, 可表示物体在 Ox 轴上的简谐运动。



$a_n = A\omega^2$

$a = A\omega^2 \cos(\omega t + \varphi + \pi)$

$= -A\omega^2 \cos(\omega t + \varphi)$

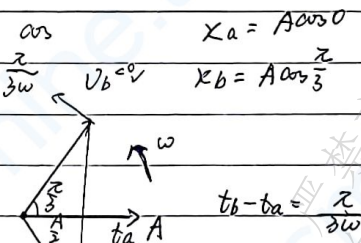
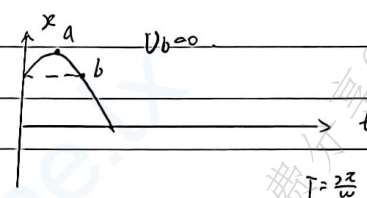


1. 对同一简谐运动, 相位差可以给出两运动状态变化所需时间。

相位差 $x_1 = A \cos(\omega t_1 + \varphi)$, $x_2 = A \cos(\omega t_2 + \varphi)$

$\Delta \varphi = \omega(t_2 - t_1) = \omega \Delta t$

$\Delta t = \frac{\Delta \varphi}{\omega}$

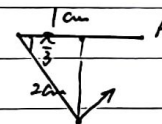


2. 两个同频率的简谐运动, 相位差表示它们时间步调的差别 (p9)

$x_1 = A_1 \cos(\omega t + \varphi_1)$ $x_2 = A_2 \cos(\omega t + \varphi_2)$

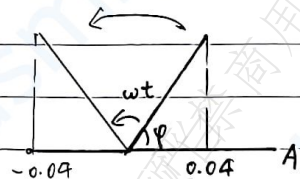
$\Delta \varphi = \varphi_2 - \varphi_1$

e.g. 1 $t=0$, $x = 1 \text{ cm}$ 自 Ox 轴正向运动 $A = 2 \text{ cm}$, $\omega = 4\pi \text{ rad/s}$



$\therefore x = 0.02 \cos(4\pi t - \frac{\pi}{3}) \text{ (m)}$

e.g.2. $A=0.08\text{m}$ $T=4\text{s}$ $t=0$ $x=0.04\text{m}$ 向OX轴正向



$$\varphi = \frac{\pi}{3}$$

$$\omega t = \frac{\pi}{3}$$

$$\omega = \frac{2\pi}{T} = \frac{\pi}{2}$$

$$\therefore t = \frac{2}{3}\text{s}$$

e.g.3. $x-t$ 曲线



(1) $x = 0.1 \cos \omega t$

$$\omega = \frac{\frac{\pi}{2} + \frac{\pi}{2}}{0.4} = \frac{\pi}{0.4} = \frac{5\pi}{2}$$

$$x = 0.1 \cos(\frac{5\pi}{2}t - \frac{\pi}{3})$$

(2) $\varphi_p = 0$

(3) $\Delta \varphi' = \omega \Delta t'$ $\Delta t' = \frac{\Delta \varphi'}{\omega} = 1.6\text{s}$

简谐运动 确定运动状态: 由x和v结合确定.

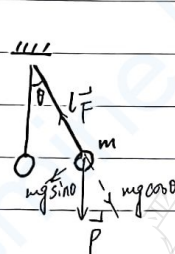
或相位 (不可直接观测)

↓ 引入路程变量

变量在x轴的位移速度加速度上的平均分量
与简谐运动一一对应.

9-3 单摆和复摆

一. 单摆



↑ 受力分析.

$$F > mg \cos \theta$$

$$F_t = -mg \sin \theta \approx -mg \theta$$

$$a_t = l \frac{d^2 \theta}{dt^2}$$

$$-g \theta = l \frac{d^2 \theta}{dt^2}$$

$$g \theta = \frac{d^2 \theta}{dt^2}$$

$$\frac{d^2 \theta}{dt^2} = -\omega^2 \theta$$

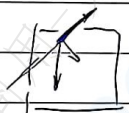
$$\omega = \sqrt{\frac{g}{l}}$$

特征方程.

$$\theta = \theta_m \cos(\omega t + \varphi)$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$\vec{M} = \vec{r} \times \vec{F}$ 力矩垂直纸面向里.



$$M = -mgl \sin \theta = J \alpha = ml^2 \frac{d^2 \theta}{dt^2}$$

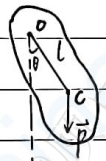
$$\alpha = \frac{d^2 \theta}{dt^2} = -\frac{g}{l} \theta$$

$$\omega^2 = \frac{g}{l} \therefore \frac{d^2 \theta}{dt^2} = -\omega^2 \theta$$

(p5)

二. 复摆

转动正向



$$\vec{M} = \vec{r} \times \vec{p}$$

$$M = -mgl \sin \theta = J \alpha = J \frac{d^2 \theta}{dt^2}$$

$$-mg l \theta = J \frac{d^2 \theta}{dt^2}$$

$$\omega^2 = \frac{mgl}{J}$$

$$\therefore \frac{d^2 \theta}{dt^2} = -\omega^2 \theta \rightarrow \text{特征方程}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{J}{mgl}}$$

$$\theta = \theta_m \cos(\omega t + \varphi)$$

三. 简谐运动方程和特征

线性回复力 $F = -kx$ 平衡位置 $x=0$

本资料来源于网络. 如遇侵权, 请联系删除. 并联系: gubaibai.ovo@gmail.com

2. 动力学方程 $a = \frac{d^2x}{dt^2} = -\omega^2 x$

3. 运动方程 $x = A \cos(\omega t + \varphi)$

三种模型

弹簧振子 $\omega = \sqrt{\frac{k}{m}} \quad T = 2\pi \sqrt{\frac{m}{k}}$

单摆 $\omega = \sqrt{\frac{g}{l}} \quad T = 2\pi \sqrt{\frac{l}{g}}$

复摆 $\omega = \sqrt{\frac{mgl}{I}} \quad T = 2\pi \sqrt{\frac{I}{mgl}}$

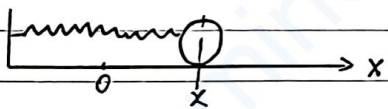
e.g.1 $F = -x^2 x$ 非简谐运动

$ma = -x^2 x$

$a = -(\frac{x}{\sqrt{m}})^2 x \quad \omega = \frac{x}{\sqrt{m}} \quad T = \frac{2\pi}{\omega} = 2\sqrt{m}$

9.4 简谐振动的能量

1. 动能



$E_k = \frac{1}{2} m v^2 = \frac{1}{2} m [-\omega A \sin(\omega t + \varphi)]^2 = \frac{1}{2} m \omega^2 A^2 \sin^2(\omega t + \varphi) \quad \omega^2 = \frac{k}{m}$

$E_k = \frac{1}{2} k A^2 \sin^2(\omega t + \varphi) \quad T = \frac{T_0}{2}$

2. 势能

$E_p = \frac{1}{2} k x^2 = \frac{1}{2} k A^2 \cos^2(\omega t + \varphi)$

3. 机械能 $E = E_k + E_p = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} k A^2 = \frac{1}{2} m v_{\max}^2$

↓ 位移最大处
↓ 原处处动能

线性回复力是保守力，作简谐振动的系统机械能守恒

简谐振动能能量图 P14

简谐运动能量守恒，振幅不变

由能量守恒推导简谐运动微分方程

$E = \frac{1}{2} k x^2 + \frac{1}{2} m v^2 = C$

$\frac{dE}{dt} = \frac{d}{dt} (\frac{1}{2} k x^2 + \frac{1}{2} m v^2) = 0$ (时间平移不变性)

$m \times \frac{dv}{dt} + k \times \frac{dx}{dt} = 0$

$\frac{d^2x}{dt^2} + \frac{k}{m} x = 0$

e.g.1 $\frac{1}{16} E_p \leftarrow \frac{1}{4} A$
 $E_k = \frac{15}{16} E \quad A$
 $E = E_p$

$\vec{A}_1, \vec{A}_2$ 矢量叠加!

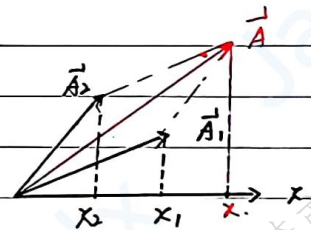
9.5 简谐运动合成

1. 两个同方向同频率简谐运动合成

$x_1 = A_1 \cos(\omega t + \varphi_1)$

$x_2 = A_2 \cos(\omega t + \varphi_2)$

$\Delta\varphi = \varphi_2 - \varphi_1 = C$



$A = \sqrt{A_1^2 + A_2^2 + 2A_1 A_2 \cos(\varphi_2 - \varphi_1)}$

$\tan\varphi = \frac{A_1 \sin\varphi_1 + A_2 \sin\varphi_2}{A_1 \cos\varphi_1 + A_2 \cos\varphi_2}$

$x = A \cos(\omega t + \varphi)$

$\Delta\varphi = \varphi_2 - \varphi_1 = 2k\pi \rightarrow x = (A_1 + A_2) \cos(\omega t + \varphi)$ 振动加强

$\Delta\varphi = \varphi_2 - \varphi_1 = (2k+1)\pi \rightarrow x = |A_1 - A_2| \cos(\omega t + \varphi)$ 振动减弱

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第十章 波动

10-1 机械波的几个概念

一. 机械波的形成

1. 波源：作机械振动的物体。

2. 介质

波是运动状态的传播，介质的质点并不随波传播。

二. 横波和纵波

1. 横波

质点振动方向

波传播方向

行波、光波

波峰/谷

2. 纵波

振动方向

波传播方向

弹簧波、声波

波疏/密区

三. (P46)

1. 波长 λ

2. 波速 u

3. 周期 T

4. 频率 ν

决定于介质的性质

决定于波源的性质

$$\nu = T^{-1}$$

$$u = \frac{\lambda}{T} = \lambda \nu$$

$$\lambda = \frac{u}{\nu} = uT$$

e.g. 1 (P47) $\lambda = uT = \frac{u}{\nu}$

在空气中:

$$\lambda_1 = \frac{340}{200} = 1.7 \text{ m}$$

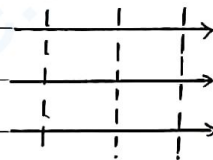
$$\lambda_2 = \frac{340}{2000} = 0.17 \text{ m}$$

在水中:

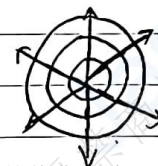
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四. 波线 波面 波前 P48.

1. 平面波



2. 球面波

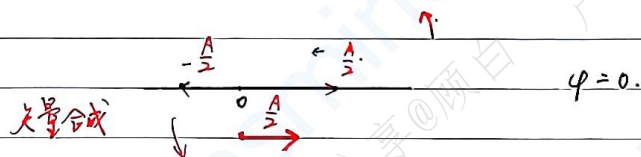
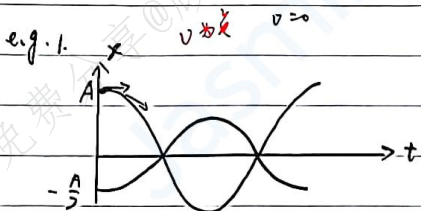


$$\varphi_1 =$$

$$\varphi = \varphi_2 - \varphi_1 =$$

T 相同

同方向同频率



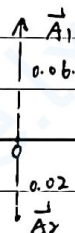
学习指导 P144T8.

e.g. 2. 同方向. 频率 $\varphi_1 = \frac{\pi}{2}$

$$x_1 = 0.06 \cos(5t + \frac{\pi}{2})$$

$$x_2 = 0.02 \cos(5t - \frac{\pi}{2})$$

$$\varphi_2 = -\frac{\pi}{2}$$



$$\vec{A} = 0.04$$

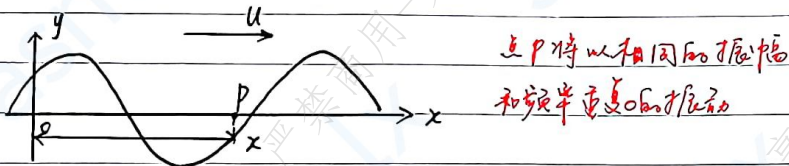
$$x = 0.04 \cos(5t + \frac{\pi}{2})$$

$$A = 0.04 \text{ m}$$

$$\varphi = \frac{\pi}{2} \text{ rad}$$

x 处的质元在任意时刻 t 的位移 $y \rightarrow y = y(x, t)$

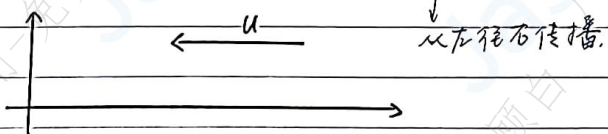
10-2 平面简谐波的波动方程 $y(x, t)$ 波动方程和波动函数 $y(x, t)$ 为坐标原点



坐标原点 O 处振动方程为: $y_0 = A \cos(\omega t + \varphi)$

设 P 坐标为 x , P 振动落后 O 的时间: $\Delta t = \frac{x}{u}$

$$\begin{aligned} \therefore P \text{ 的振动方程 } y_P &= y_0(t - \Delta t) \\ &= A \cos[\omega(t - \Delta t) + \varphi] \\ &= A \cos[\omega(t - \frac{x}{u}) + \varphi] = y(x, t) \end{aligned}$$



P 振动超前 O 振动时间: $\Delta t = \frac{x}{u}$

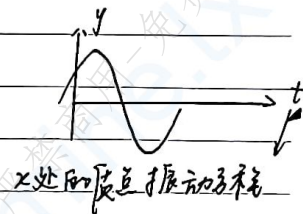
$$\begin{aligned} y &= y_0(t + \Delta t) = A \cos[\omega(t + \frac{x}{u}) + \varphi] \\ &\quad \downarrow \text{对应} \\ &\quad \text{从左向右传播} \end{aligned}$$

$$\begin{aligned} y &= y_0(t - \Delta t) = A \cos[\omega(t - \frac{x}{u}) + \varphi] = A \cos[2\pi(\frac{t}{T} - \frac{x}{\lambda}) + \varphi] \\ &= A \cos[\omega t - \frac{2\pi x}{\lambda} + \varphi] \end{aligned}$$

2. 波动方程的物理含义

x 是 t 变化 $y = A \cos(\omega t - \frac{2\pi x}{\lambda} + \varphi)$

$$\text{令 } \varphi' = \varphi - \frac{2\pi x}{\lambda} \quad y = A \cos(\omega t + \varphi')$$

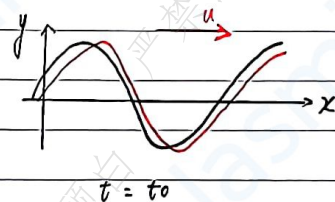


2. t 是 x 变化

$$\text{令 } \varphi'' = \omega t + \varphi = C$$

$$\text{则 } y = A \cos(-\frac{2\pi x}{\lambda} + \varphi'')$$

t 时刻 y 波传播方向上各质点的位移, 即 t 时刻的波形图。



注意: 此种情况的前提为

$$y_A = A \cos(-\frac{2\pi x_A}{\lambda} + \varphi'') \quad y_B = A \cos(-\frac{2\pi x_B}{\lambda} + \varphi'')$$

$$\varphi_A - \varphi_B = \frac{2\pi(x_B - x_A)}{\lambda} = -\frac{2\pi}{\lambda}(x_A - x_B)$$

$$x_A > x_B \quad \varphi_A < \varphi_B \quad A \text{ 落后于 } B \text{ 振动}$$

$$x_A < x_B \quad \varphi_A > \varphi_B \quad A \text{ 超前于 } B \text{ 振动}$$

波速的方向是

从左向右传播

3. x, t 都变 P52

$$y = 2 \cos[4(t + \frac{7}{4}x)]$$

$$u = -\frac{4}{7} \text{ m/s}$$

$$T = \frac{2\pi}{\omega} = \frac{\pi}{2} \text{ s}$$

eg. 1 波动方程 $y = 2 \cos(4t + 7x) \text{ m}$ 求 T, λ, u

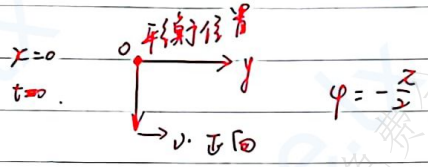
$$\lambda = uT = \frac{4}{7} \times \frac{\pi}{2} = \frac{2\pi}{7} \text{ m}$$

$$\omega = 4 \text{ rad/s}$$

$$T = \frac{2\pi}{\omega} = \frac{\pi}{2} \text{ s}$$

波函数 $y = A \cos(\omega t + \varphi)$
 波速 $x=0$
 质点的运动 y 振动的位移
 e.g. 2. (p53). 在 $t=0$ 时坐标原点的质点在平衡位置沿 oy 正向运动

(1) 波速数
 $u = \frac{\lambda}{T} = 1 \text{ m/s}$ $\omega = \frac{2\pi}{T} = \pi$
 $y = \cos[\pi(t-x) + \varphi]$
 $y(0,0) = 0$



注意明确角为何 φ 即为坐标原点的初相
 在推导波函数时, 我们以坐标原点为基准, 利用振动的位移推波函数
 $y = \cos[\pi(\frac{1}{2} - x) - \frac{\pi}{2}]$ (m).

(2) $x=0.5 \text{ m}$
 所以波函数中的 φ 与基准点的 φ 相同.
 利用旋转矢量求出质点振动的初相, 即求出波函数中的 φ .
 $y = \cos(\pi t - \pi) = -\cos \pi t$ 振动的位移



(3) $t=1 \text{ s}$ 波形图
 $y = \cos(\frac{\pi}{2} - \pi x) = \sin \pi x$ $t=1 \text{ s}$ 时的波形图

e.g. 3. (1) 波速 u
 $y_{0A} = 3 \times 10^{-2} \cos(4\pi t)$

(1) 以 A 为原点, $y_{0A} = 0.03 \cos 4\pi t$
 $y_A = 0.03 \cos[4\pi(t - \frac{x}{20})]$

(2) $x_B = -5 \text{ m}$ B 的振动方程 $y_{0B} = 0.03 \cos(4\pi t + \pi)$
 利用 A 的振动方程求
 直接利用 B 超前 A 求出了波数 (以 A 为 0 点)
 $y_B = 0.03 \cos[4\pi(t - \frac{x}{20}) + \pi]$
 求 B 的振动方程
 波数即求波数 (以 B 为 0 点)

波速 y_{0A} 波速 y_{0B} 波速 y_{0C}
 是 x_c, x_d 以 y 轴为原点. 选 A 为坐标原点和选 B 为坐标原点, x_c 和 x_d 的表示法不同

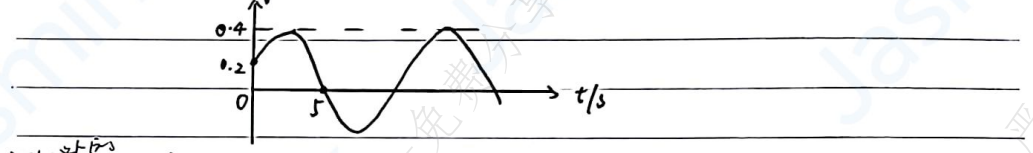
以 A 为坐标原点, $x_c = -13 \text{ m}$ 代入 y_A
 $y_{0C} = 0.03 \cos[4\pi(t + \frac{13}{20})]$
 $x_D = 9 \text{ m}$ 代入 y_A
 $y_{0D} = 0.03 \cos[4\pi(t - \frac{9}{20})]$

求 BC 和 CD 两质点间的相位差. 波数
 即 t 相同时, 不同 x 之间的相位差
 (4) $T = \frac{2\pi}{4\pi} = 0.5 \text{ s}$ $u = 20 \text{ m/s}$ $\lambda = uT = 10 \text{ m}$ 差异

$\varphi_B - \varphi_C = -\frac{2\pi}{\lambda}(x_B - x_C) = -\frac{16\pi}{10} = -1.6\pi$ BC 相隔 0.8 个波长

$\varphi_C - \varphi_D = -\frac{2\pi}{\lambda}(x_C - x_D) = 4.4\pi$ CD 相隔 2.2 个波长

e.g. 4. 一平面简谐波, 波长为 12 m , 沿 x 负方向传播. 图示为 $x=1 \text{ m}$ 处质点振动曲线. 求波函数



$x=1 \text{ m}$ 处的质点振动方程 $y = 0.4 \cos(\frac{\pi}{6}t - \frac{\pi}{3})$ (m).
 $\omega = \frac{\pi + \frac{\pi}{3}}{1} = \frac{\pi}{6}$

$x=1 \text{ m}$ 处的质点振动方程为 $y = 0.4 \cos(\frac{\pi}{6}t - \frac{\pi}{3})$ (m).
 $u = \frac{\lambda}{T} = \frac{12 \times \frac{\pi}{6}}{2\pi} = 1 \text{ m/s}$ $\Delta t = \frac{1}{1} = 1 \text{ s}$ 原点处落后 1 s

原点处的振动方程 $y = 0.4 \cos[\frac{\pi}{6}(t-1) - \frac{\pi}{3}] = 0.4 \cos[\frac{\pi}{6}t - \frac{\pi}{2}]$
 波函数 $y = 0.4 \cos[\frac{\pi}{6}(t + \frac{x}{1}) - \frac{\pi}{2}] = 0.4 \sin[\frac{\pi}{6}(x+t)]$

10-3 波的能量 能流密度

一 振动数 (取 $\phi_0 = 0$) $y = A \cos[\omega(t - \frac{x}{u})]$

$$\frac{\partial^2 y}{\partial t^2} = -\omega^2 A \cos[\omega(t - \frac{x}{u})]$$

$$\frac{\partial^2 y}{\partial x^2} = -\frac{\omega^2}{u^2} A \cos[\omega(t - \frac{x}{u})]$$

$\Downarrow$

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{u^2} \frac{\partial^2 y}{\partial t^2}$$

波速平方为二

二 波动能量的传播

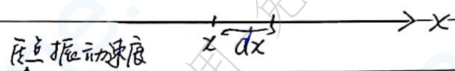
1. 波的能量

介质质点运动 $\rightarrow$ 动能 E_k

介质形变势能 $\rightarrow W_p$

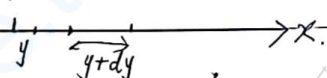
以横波传播纵波分析

振动动能



$$y = A \cos[\omega(t - \frac{x}{u})] \quad v = -A\omega \sin[\omega(t - \frac{x}{u})] \rightarrow \text{波形传播速度}$$

$$dW_k = \frac{1}{2}(\rho \Delta x) v^2 = \frac{1}{2}(\rho \Delta x) \omega^2 A^2 \sin^2[\omega(t - \frac{x}{u})]$$



弹性势能

$$dW_p = \frac{1}{2}k(dy)^2$$

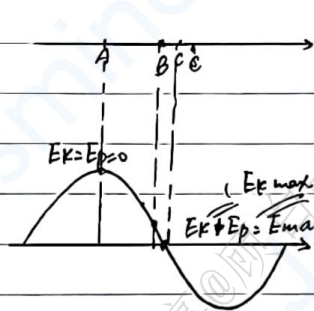
$$F = k \Delta y \frac{ES}{L} \frac{dy}{dx}$$

$\frac{F}{S} = E \left(\frac{dy}{dx}\right)$ 应变
应力 $\rightarrow$ 横截面积

$$k = \frac{ES}{\Delta x} \quad dW_p = \frac{1}{2}k(dy)^2 = \frac{1}{2}ES \frac{dx}{\Delta x} \left(\frac{dy}{dx}\right)^2 = \frac{1}{2}E \Delta x \left(\frac{dy}{dx}\right)^2 = \frac{1}{2}\rho u^2 \Delta x \left(\frac{dy}{dx}\right)^2$$

$$dW_p = \frac{1}{2}\rho \Delta x A^2 \omega^2 \sin^2[\omega(t - \frac{x}{u})] = dW_k$$

$$\therefore dW = dW_k + dW_p = \rho \Delta x A^2 \omega^2 \sin^2[\omega(t - \frac{x}{u})]$$



a. 波动传播介质中, E_k, E_p, E 变化同相位

E_k, E_p 不相等 $\left\{ \begin{array}{l} \text{平衡位置: } E_k, E_p \text{ 均 } E \text{ 最大} \\ \text{位移最大: } E_k, E_p \text{ 均为 } 0 \end{array} \right.$

任一体积元

b. 机械能不守恒, ω 为常量

体积元从后面介质中获得能量

波动是能量传播的载体, 又不断把能量传给前面的

2. 能量密度

$$\bar{w} = \frac{dW}{dV} = \rho A^2 \omega^2 \sin^2[\omega(t - \frac{x}{u})]$$

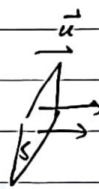
3. 平均能量密度

$$\bar{w} = \frac{1}{T} \int_0^T \bar{w} dt = \frac{1}{2} \rho A^2 \omega^2$$

三. 能流和能流密度

单位时间内垂直通过某面积的能量

$$P = \frac{dW}{dt} = \bar{w} S u$$



$$\text{平均能流: } \bar{P} = \bar{w} S u = \frac{1}{2} \rho A^2 \omega^2 u S$$

$$\text{能流密度: } I = \frac{\bar{P}}{S} = \bar{w} u = \frac{1}{2} \rho A^2 \omega^2 u$$

例题

e.g. 1

10-4 惠更斯原理 衍射干涉

↓ p58

- 波动在各向同性介质传播, 波阵面几何形状不变
- 波动在不均匀介质或各向异性介质, 几何形状, 传播方向改变
- Huygens 原理并未说明波强度分布, 只能用于解决波的传播方向问题

障碍物

二. 波的衍射 (宽度小于波长) p59.

三. 波的干涉

1. 波的叠加原理 p60.

(1) 波传播的独立性

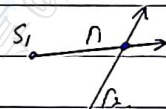
(2) 波的叠加性

2. 波的干涉条件: 频率相同, 振动方向相同, 相位差恒定 (两列波)

干涉现象: 某些点振动始终加强, 某些点振动始终减弱.

3. 定量讨论 频率相同:

$$\begin{cases} y_1 = A_1 \cos(\omega t + \varphi_1) \\ y_2 = A_2 \cos(\omega t + \varphi_2) \end{cases}$$



P 点的两个分振动

$$\begin{cases} y_{1P} = A_1 \cos(\omega t + \varphi_1 - \frac{2\pi r_1}{\lambda}) \\ y_{2P} = A_2 \cos(\omega t + \varphi_2 - \frac{2\pi r_2}{\lambda}) \end{cases} \quad \Delta\varphi = \varphi_2 - \varphi_1 - 2\pi \frac{r_2 - r_1}{\lambda} \text{ 定值}$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos\varphi}$$

$$\Delta\varphi = 2k\pi \quad (k \in \mathbb{Z}) \text{ 时} \quad A = A_{\max} = A_1 + A_2 \quad \text{干涉加强}$$

$$\Delta\varphi = (2k+1)\pi \quad (k \in \mathbb{Z}) \text{ 时} \quad A = A_{\min} = |A_1 - A_2| \quad \text{干涉减弱}$$

如

$$\text{若 } \varphi_2 = \varphi_1, \text{ 即 } S_1, S_2 \text{ 同相位} \quad \Delta\varphi = \frac{2\pi}{\lambda}(r_1 - r_2) = \frac{2\pi}{\lambda}\delta = \begin{cases} 2k\pi \\ (2k+1)\pi \end{cases}$$

$$\delta = r_1 - r_2 = k\lambda \quad \text{加强}$$

$$\delta = r_1 - r_2 = (2k+1)\frac{\lambda}{2} \quad \text{减弱}$$

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$$\Delta\varphi = \pi, \varphi_A - \varphi_B$$

$$A = 5 \text{ cm}$$

eq.

15

25

A

B

20

$$r_{PB} = r_{PB1} = 25$$

$$r_A = r_{PA} = 15 \text{ cm}$$

$$r_B = r_{AB} = 20 \text{ cm}$$

$$r_B - r_A = 10 \text{ cm}$$

$$\Delta\varphi = \varphi_B - \varphi_A - 2\pi \frac{r_B - r_A}{\lambda} \quad \lambda = \frac{u}{f} = \frac{u}{\nu} = \frac{10}{100} = 0.1 \text{ m}$$

$$\Delta\varphi = \pi - 2\pi \frac{10}{0.1} = -199\pi \quad A = |A_1 - A_2| = 0$$

干涉减弱.

10-5 驻波 (standing wave). 波形不传播.

条件: 两列频率相同. 振幅. 振动方向相同.

2. 驻波方程:

$$y_1 = A \cos(\omega t - \frac{2\pi}{\lambda} x)$$

$$y_2 = A \cos(\omega t + \frac{2\pi}{\lambda} x)$$

↓

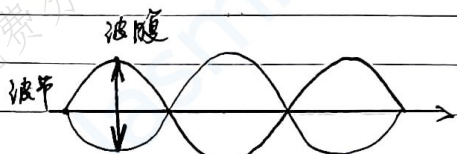
$$y = y_1 + y_2 = A \cos(\omega t - \frac{2\pi}{\lambda} x) + A \cos(\omega t + \frac{2\pi}{\lambda} x)$$

$$y = 2A \cos \frac{2\pi}{\lambda} x \cos \omega t$$

$$A' = |2A \cos \frac{2\pi}{\lambda} x|$$

a. $\cos \frac{2\pi}{\lambda} x = 0 \quad x = (2k+1) \cdot \frac{\lambda}{4} \quad k \in \mathbb{Z} \quad A' = 0$ 为波节

b. $\cos \frac{2\pi}{\lambda} x = \pm 1 \quad x = 2k \cdot \frac{\lambda}{4} \quad k \in \mathbb{Z} \quad A' = 2A$ 为波腹.



相邻波节(波腹)为 $\frac{\lambda}{2}$.

2. 相位分布

$$y = 2A \cos \frac{2\pi}{\lambda} x \cos \omega t$$

$$x \in (-\frac{\lambda}{4}, \frac{\lambda}{4}), \cos \frac{2\pi}{\lambda} x > 0$$

$$y = |2A \cos \frac{2\pi}{\lambda} x| \cos \omega t$$

相邻波节: 同各点振动相位相同.

$$x \in (\frac{\lambda}{4}, \frac{3\lambda}{4}), \cos \frac{2\pi}{\lambda} x < 0$$

$$y = -|2A \cos \frac{2\pi}{\lambda} x| \cos \omega t = |2A \cos \frac{2\pi}{\lambda} x| \cos(\omega t + \pi)$$

-波节: 两侧各点振动相位相反.

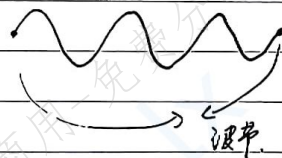
疏→密.

相位跃变 (半波损失).

形成波节

波从波疏介质到波密介质, 波反射到波疏介质

反射波在分界处产生 π 的相位跃变.



四. 驻波能量, p68.

五. 振动的简正模式.

$$l = n \cdot \frac{\lambda}{2} \quad \lambda_n = \frac{2l}{n} \quad \nu_n = n \cdot \frac{\nu}{2}, n = 1, 2, \dots$$

一端固定 一端自由

波腹.

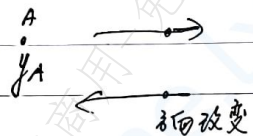
波节

端点

(p70).

eg. $y_1 = 0.001 \cos[200\pi(t - \frac{x}{200})] \text{ m}$

$L = 2.25 \text{ m}$



$y_2 = 10^{-3} \cos[200\pi(t + \frac{x}{200}) + \varphi_0]$
方向改变.

$y_{1A} = 0.001 \cos[200\pi(t - \frac{2.25}{200}) + \pi]$
相位改变.

$y_{2A} = 10^{-3} \cos[200\pi(t + \frac{2.25}{200}) + \varphi_0]$

$y_{1A} = y_{2A}$
 $2.25\pi + \varphi_0 = -1.75\pi + 2k\pi \Rightarrow \begin{cases} 0.75\pi, & k=1 \\ 2.75\pi, & k=2 \end{cases}$
 $\varphi_0 = 0.5\pi$

$\therefore y_2 = 10^{-3} \cos[200\pi(t + \frac{x}{200}) + \frac{\pi}{2}]$

(2) $y = y_1 + y_2 = 2 \times 10^{-3} \cos(\pi x + \frac{\pi}{4}) \cos(200\pi t + \frac{\pi}{4})$

(3) 波节 $\cos(\pi x + \frac{\pi}{4}) = 0$ $\pi x + \frac{\pi}{4} = k\pi + \frac{\pi}{2}$ $k \in \mathbb{Z}$
 $x = k + \frac{1}{4}$ $x = 0.25 \text{ m}, 1.25 \text{ m}, 2.25 \text{ m}.$

波腹 $\cos(\pi x + \frac{\pi}{4}) = \pm 1$ $\pi x + \frac{\pi}{4} = k\pi$ $x = k - \frac{1}{4}$ $x = 0.75 \text{ m}, 1.75 \text{ m}.$

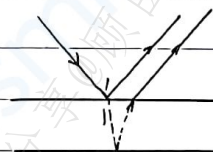
第十一章 光学

11-1 相干光 (coherent light).

1. 一个光源只能发出频率一定, 振动方向一定, 的光

相干光的产生. 振幅分割法.

波阵面分割法.



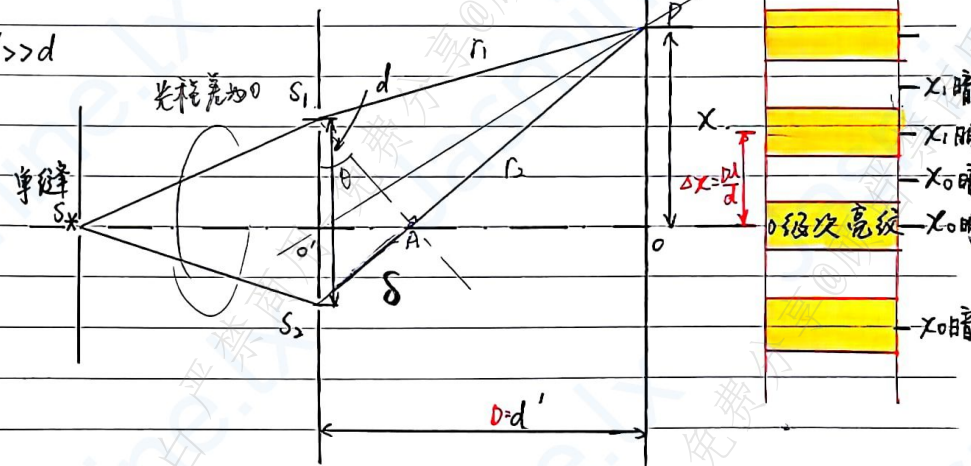
薄膜干涉.

同波阵面上振动状态相同

光程差 $\Delta r = \frac{2\pi}{\lambda} \delta$ 光程差 $\delta = \frac{d}{D} \cdot x$ 相位差 $\Delta \phi = \frac{2\pi}{\lambda} \delta$ 相位差 $\Delta \phi = \frac{2\pi}{\lambda} \delta$

11-2 杨氏双缝干涉.

$d' \gg d$



$\Delta r = r_2 - r_1$ $d' \gg d$ $\delta = S_2 A = r_2 - r_1 = \Delta r = d \sin \theta$

$\angle P O' O \approx \theta$ $\sin \theta \approx \tan \theta = \frac{x}{d'}$

$\therefore$ 光程差 $\Delta r = d \sin \theta = d \frac{x}{d'}$

光程差

$\delta = \frac{d}{D} \cdot x$

近似 $S_2 A \approx r_2 - r_1$ $\angle P O' O \approx \theta$ $\sin \theta \approx \tan \theta$

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成立关键: $d' \gg d$

$\delta = \begin{cases} \pm k \cdot \lambda & k=0, 1, 2, 3 \\ \pm (2k+1) \cdot \frac{\lambda}{2} & k=0, 1, 2, 3 \end{cases}$
 $k=0$ 0级暗纹
 $\delta = d \cdot \frac{x}{d'} = \begin{cases} k \cdot \lambda & \text{加强} \\ (2k+1) \cdot \frac{\lambda}{2} & \text{减弱} \end{cases}$
 如果求k级暗纹，取 $k=4$ 代入。

$x = \begin{cases} k \cdot \frac{d'}{d} \cdot \lambda & k \text{ 级明纹} \\ (2k+1) \cdot \frac{d'}{d} \cdot \frac{\lambda}{2} & k \text{ 级暗纹} \end{cases}$
 $x_{k+1} - x_k = \Delta x = [(2k+3) - (2k+1)] \cdot \frac{d'}{d} \cdot \frac{\lambda}{2}$
 (明)条纹间距：
 相邻暗条纹中间的距离 $= \frac{d'}{d} \cdot \lambda$
 $181 < d$
 $k \cdot \lambda \leq d$
 $k < \frac{d}{\lambda}$
 $k_{\max} = [\frac{d}{\lambda}]$
 向下取整

$\lambda_{\text{红}} > \lambda_{\text{紫}}$
 eg $\lambda = 589.3 \text{ nm}$ $d' = 800 \text{ mm}$
 1) $d = 1 \text{ mm}$
 $\Delta x = \frac{d'}{d} \cdot \lambda = 800 \times 589.3 \times 10^{-9} = 4.7 \times 10^{-4} \text{ m} = 0.47 \text{ mm}$

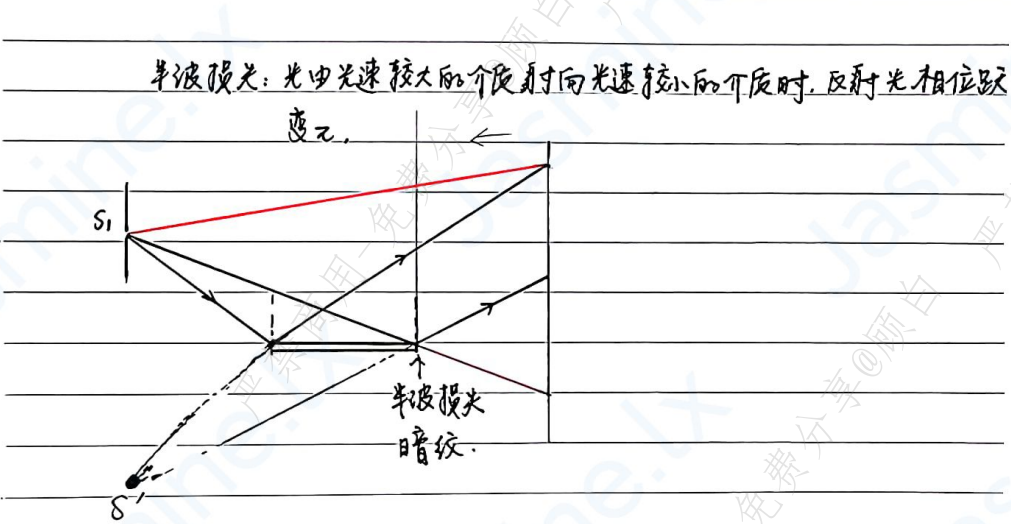
2) $\Delta x' = \frac{1}{10} \Delta x = 0.047 \text{ mm}$
 $d' \gg d$
 约三个数量级。
 eg. $d = 0.2 \text{ mm}$ $d' = 1 \text{ m}$
 1) $4 \cdot \frac{d'}{d} \cdot \lambda - \frac{d'}{d} \cdot \lambda = 3 \frac{d'}{d} \cdot \lambda = 3 \Delta x = 7.5$
 $\lambda = 7.5 \times \frac{1}{3} \times \frac{d}{d'} = 7.5 \times \frac{1}{3} \times \frac{0.2 \text{ mm}}{1 \text{ m}} = 500 \text{ nm}$ 绿。

2) $\Delta x = \frac{1}{2} \frac{d'}{d} \cdot \lambda = 1.5 \text{ mm}$

二. 缝宽对干涉条纹的影响 空间相干性
 光源

光的时间相干性 波程差 $<$ 波列长度
 $500 \times 10^{-9} \text{ m}$
 $\Delta r = c \cdot \Delta t \leq \Delta l$
 $\Delta t \leq \frac{\Delta l}{c}$
 $\Delta l = 5000 \times 10^{-9} \text{ m}$
 $= 5000 \text{ nm}$
 $= 5 \times 10^{-3} \text{ mm}$

四. 劳埃德镜 半波损失



// 计

光的相干性

1. 两列光波相遇发生干涉的必要条件:

- ① 振动方向相同.
- ② 振动频率相同.
- ③ 相位差恒定.

普通光源获得相干光:

- ① 分波阵面法
- ② 振幅分割法

1. 光程: $\Delta n \cdot r \rightarrow$ 光走过的几何路程.

折射率

光在介质中的波长

光在真空中的波长.

$$v = \frac{c}{n}$$

$$\lambda' = \frac{\lambda}{n}$$

$$\nu = \frac{c}{\lambda}$$

2. 光程差: $\delta = n_2 r_2 - n_1 r_1$

$$E_1 = E_{01} \cos(\omega t - k_1 r_1 + \varphi_0)$$

$$E_2 = E_{02} \cos(\omega t - k_2 r_2 + \varphi_0)$$

$$k = \frac{2\pi}{\lambda}$$

$$\Delta\varphi = \varphi_1 - \varphi_2 = k_2 r_2 - k_1 r_1$$

$$= \frac{2\pi}{\lambda_2} r_2 - \frac{2\pi}{\lambda_1} r_1$$

$$= 2\pi \left(\frac{r_2}{\lambda_2} - \frac{r_1}{\lambda_1} \right)$$

$$\Delta\varphi = 2\pi \left(\frac{n_2 r_2}{n_2 \lambda_2} - \frac{n_1 r_1}{n_1 \lambda_1} \right)$$

$$= 2\pi \cdot \frac{\delta}{\lambda}$$

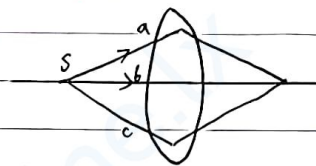
相位差与光程差的关系.

$$\Delta\varphi = \frac{2\pi}{\lambda} \cdot \delta$$

相位



使用透镜不会产生附加光程差.



a, b, c 到 S 到 S' 光程相同.

$$1^\circ \Delta\varphi = \frac{2\pi}{\lambda} \delta = 2k\pi$$

相位

$$\Delta\varphi = \frac{2\pi}{\lambda} \delta$$

光程差

由相位对应的干涉情况. 推光程差对应的干涉情况.

$$\delta = \pm 2k \cdot \frac{\lambda}{2} \quad k=0, 1, 2, \dots \quad \text{相加强}$$

$$2^\circ \Delta\varphi = \frac{2\pi}{\lambda} \delta = (2k+1)\pi$$

$$\delta = \pm (2k+1) \cdot \frac{\lambda}{2} \quad k=0, 1, 2, \dots \quad \text{相干相消}$$

杨氏双缝干涉

双缝距

相位差

$$\Delta\varphi = \frac{2\pi}{\lambda} \cdot \delta$$

光程差

$$\delta = \frac{d}{D} \cdot x$$

双缝实验参数.

$$\delta = \frac{d}{D} \cdot x = \begin{cases} k \cdot \lambda \\ (2k+1) \frac{\lambda}{2} \end{cases}$$

$$x = \begin{cases} k \cdot \frac{D}{d} \cdot \lambda \\ (2k+1) \frac{D}{d} \cdot \frac{\lambda}{2} \end{cases}$$

$$\Delta x = \frac{D}{d} \cdot \lambda$$

杨氏双缝干涉例题.

题型1

例1. 单色光照射到 $d=0.2\text{mm}$ 的双缝上. 双缝到屏 $D=1\text{m}$.

(1). 从第1级明纹到同时第4级明纹间的距离为 7.5mm , 求单色光波长.

(2). 若入射光的波长为 600nm , 中央明纹中心最近的暗纹中心距离是多少?

明纹

$$\text{解: (1). } x = k \cdot \frac{d}{D} \lambda$$

暗纹

$$x = (k+1) \times \frac{d}{D} \cdot \lambda$$

$$= k \cdot \Delta x$$

$$\Delta x_{1,4} = 3 \Delta x = 3 \cdot \frac{D}{d} \cdot \lambda$$

$$\therefore \lambda = \frac{d \Delta x_{1,4}}{3 D}$$

$$\therefore \Delta x = 2.5\text{mm} = \frac{D}{d} \lambda$$

$$= 0.2 \times 10^{-3} \times 7.5 \times 10^{-3}$$

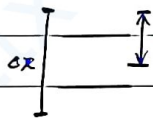
$$3 \times 1$$

$$= 0.5 \times 10^{-6} \text{m}$$

$$= 500\text{nm}$$

$$\lambda =$$

$$(2). \Delta x' = \frac{1}{2} \Delta x = \frac{1}{2} \cdot \frac{D \lambda}{d} = 1.5\text{mm}$$



$$\delta \neq 0 !!!$$

$$\delta \neq 0 !!!$$

例2. 单色光源 S_0 到 S_1 和 S_2 距离分别为 l_1 和 l_2 , 且 $l_1 - l_2 = 4\lambda$, λ 为入射波长,

双缝间距为 d . 双缝到屏为 D . 求:

(1). 零级明纹到屏幕中央 O 的距离.

(2). 正四级、负四级明纹到中央 O 的距离.

解:

求两束光的光程差 (所有光程之差).

$$\delta = (l_2 + r_2) - (l_1 + r_1)$$

$$= (r_2 - r_1) - (l_1 - l_2)$$

$$= \underbrace{r_2 - r_1}_{\delta} - 4\lambda$$

$$\delta = \frac{d}{D} \cdot x - 4\lambda$$

光程差

$$\delta = \begin{cases} \pm k \cdot \lambda, & \text{明纹 } k = 0, 1, 2, \dots, k_{\max} \\ \pm (2k+1) \frac{\lambda}{2}, & \text{暗纹 } k = 0, 1, 2, \dots, k_{\max} \end{cases}$$

$$k \cdot \lambda \leq d$$

$$k \leq \frac{d}{\lambda}$$

$$(1). \text{零级明纹 } k=0 \quad \delta=0 \quad \therefore x = \frac{4D\lambda}{d}$$

正四级

$$(2). \text{正四级 } k=4 \quad \delta = \frac{d}{D} x - 4\lambda = 4\lambda \quad x = \frac{8D\lambda}{d}$$

负四级明纹

$k=-4$

$$\delta = \frac{d}{D} x - 4\lambda = -4\lambda \quad x = 0$$

$\frac{d}{2}\lambda = \frac{d}{b} \cdot x_1$ 不要忘记两条半级暗纹.

例3. 双缝干涉中两缝间距为 0.3mm , 用单色光垂直照射双缝, 在离缝 1.2m 的屏上侧测得中央明纹一侧第5级暗纹与另一侧第5级暗纹距离为 22.78mm , 问波长为多少?

解: $d = 0.3\text{mm}$, $D = 1.2\text{m}$ $\delta = \frac{d}{D} \cdot x$ $k=0, 1, 2, \dots$

$$\delta_1 = (2k+1) \cdot \frac{\lambda}{2} = \frac{d}{D} \cdot x_1 \rightarrow x_1 = \frac{5D}{2d} \cdot \lambda$$

$$\delta_2 = -\frac{\lambda}{2} = \frac{d}{D} \cdot x_2 \rightarrow x_2 = -\frac{5D}{2d} \cdot \lambda$$

$$\Delta x = x_1 - x_2 = \frac{5D}{d} \cdot \lambda$$

$$\lambda = \frac{22.78 \times 10^{-3} \times 0.3 \times 10^{-3}}{5 \times 1.2} = 2.278 \times 10^{-7} \text{m}$$

$d = 0.3\text{mm}$, $D = 1.2\text{m}$

$$\delta = \pm (2k+1) \cdot \frac{\lambda}{2}$$

第4级暗纹 $\delta = \frac{9}{2}\lambda = \frac{d}{D} \cdot x_1$

$$(\text{第4级暗纹 } \delta = -\frac{9}{2}\lambda = \frac{d}{D} \cdot x_2)$$

中央明纹与上方第5条暗纹坐标 $x = \frac{1}{2} \times 22.78 = 11.39\text{mm}$

$$\delta = (2k+1) \cdot \frac{\lambda}{2} \Big|_{k=4} = \frac{d}{D} \cdot x$$

$$\lambda = \frac{2d}{(2k+1)D} \cdot x \Big|_{k=4} = 632.8\text{nm}$$

插玻璃片

题型2.

例1: 波长 $\lambda = 500\text{nm}$ 的单色光射在相距 $d = 2 \times 10^{-4}\text{m}$ 的双缝上, 屏到双缝的距离 $D = 2\text{m}$, 用厚度 $e = 6 \times 10^{-6}\text{m}$, $n = 1.5$ 的透明薄片覆盖上面的缝, 求:

- 零级明纹到O点的距离, 其所所在位置相当于原来的第几级明纹.
- 第三级明纹到O点的距离.

解: 光程差 $\delta =$

$$\delta = r_2 - r_1 = (D - e) \times 1 + ne =$$

$$(1) \text{ 未覆盖 } r_2 - r_1 = \frac{d}{D} \cdot x$$

$$\text{覆盖后 } \delta = r_2 - (r_1 - e) \times 1 + ne = r_2 - (r_1 + (n-1)e) = r_2 - r_1 - (n-1)e$$

$$\delta = \frac{d}{D} x - (n-1)e$$

$$\text{零级明纹 } \delta = \frac{d}{D} \cdot x - (n-1)e = k \cdot \lambda \Big|_{k=0}$$

$$\therefore x = \frac{D(n-1)e}{d} = \frac{2 \times 0.5 \times 6 \times 10^{-6}}{2 \times 10^{-4}} = 0.03\text{m}$$

$$\delta' = k \cdot \lambda = \frac{d}{D} \cdot x \quad k = \frac{dx}{D\lambda} = \frac{2 \times 10^{-4} \times 3 \times 10^{-2}}{2 \times 500 \times 10^{-9}} = 6$$

相当于原来的第6级明纹.

(2) 第三级明纹:

$$\delta = \frac{d}{D} \cdot x - (n-1)e = 3\lambda$$

$$x = \frac{[(n-1)e + 3\lambda] D}{d} = \frac{[0.5 \times 6 \times 10^{-6} + 3 \times 500 \times 10^{-9}] \times 2}{2 \times 10^{-4}} = 0.045\text{m}$$

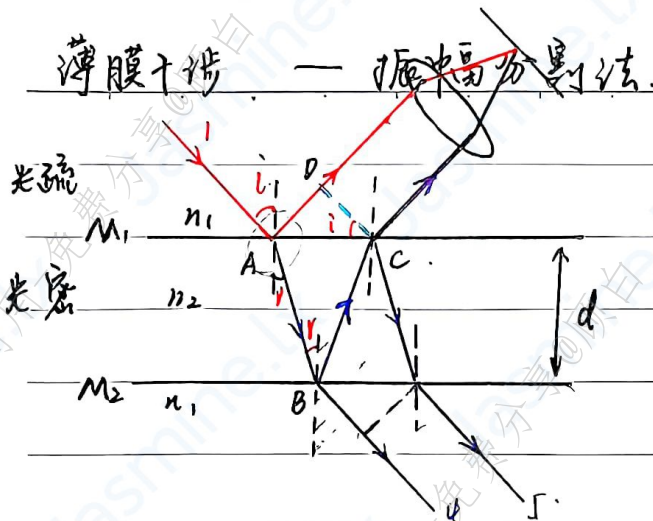


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Date _____

薄膜干涉

振幅法



$n_2 > n_1$

$$\Delta_{32} = n_2 (AB + BC) - n_1 AD + \frac{\lambda}{2} \text{ 半波损失}$$

$$AB = BC = \frac{d}{\cos r} \quad AD = AC \sin i = 2d \tan r \sin i$$

$$\therefore \Delta_{32} = \frac{2d}{\cos r} n_2 (1 - \sin^2 r) + \frac{\lambda}{2} = 2n_2 d \cos r + \frac{\lambda}{2} \quad \sin i / \sin r = n_1 / n_2$$

$$\Delta_{32} = 2d \sqrt{n_2^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2} \rightarrow \Delta r = r(i) \text{ 等倾干涉}$$

$$i = 0 \quad \Delta_{32} = 2n_2 d + \frac{\lambda}{2}$$

加强 $\Delta r = k\lambda \quad k = 1, 2, \dots$ 若 $k=0 \quad \Delta_{32} = 0 \quad \sqrt{n_2^2 - n_1^2 \sin^2 i} = 0$ 舍 $k=0$

减弱 $\Delta r = (2k+1)\frac{\lambda}{2} \quad k = 0, 1, 2, \dots$

$$\downarrow k=0 \quad \sqrt{n_2^2 - n_1^2 \sin^2 i} = 0 \text{ 有意义} \checkmark$$

$$\Delta_{4.5} = 2d \sqrt{n_2^2 - n_1^2 \sin^2 i}$$

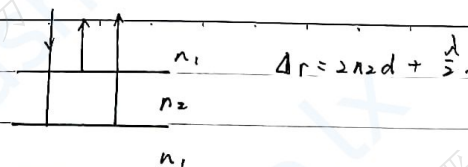
透射光和反射光干涉具有互补性 符合能量守恒定律

第 5 章

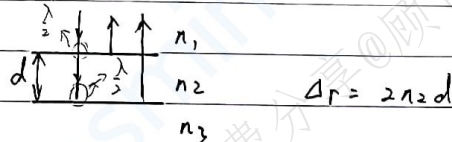
No.

Date

i=0

1. $n_2 > n_1$ 

$$\Delta r = 2n_2d + \frac{\lambda}{2}$$

2. $n_3 > n_2 > n_1$ 

$$\Delta r = 2n_2d$$

eg. 3. 1 $d = 460 \times 10^{-9} \text{ m}$

$$\lambda \in (400, 760) \text{ nm} \quad 1.2 \quad \Delta r = 2 \times 1.2 \times 460 \times 10^{-9} = 1.104 \times 10^{-6}$$

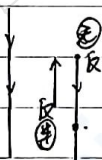
$$1.3 \quad \Delta r = 2n_1d = k\lambda$$

$$\lambda = \frac{2n_1d}{k} = \frac{1104 \text{ nm}}{k}$$

$$1104 \text{ nm}, k=1 \quad \times$$

$$552 \text{ nm}, k=2 \quad \checkmark$$

$$368 \text{ nm}, k=3 \quad \times$$



$$\Delta r = 2n_1d + \frac{\lambda}{2} = k\lambda$$

$$\lambda = \frac{2n_1d}{k - \frac{1}{2}} = \frac{1104 \text{ nm}}{k - \frac{1}{2}}$$

$$2208 \text{ nm}, k=1 \quad \times$$

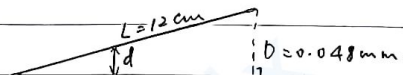
$$736 \text{ nm}, k=2 \quad \checkmark$$

$$441 \text{ nm}, k=3 \quad \checkmark$$

$$315 \text{ nm}, k=4 \quad \times$$



e.g. 1



$$\Delta r = 2n_2d + \frac{\lambda}{2} = k\lambda \quad (k=0, 1, 2, \dots)$$

半波损失.

$$2k+1 = \frac{2n_2d + \frac{\lambda}{2}}{\frac{\lambda}{2}} = \frac{4n_2d}{\lambda} + 1$$

$$2k = \frac{4n_2d}{\lambda} \quad k = \frac{2n_2d}{\lambda} \leq \frac{2n_2D}{\lambda}$$

$$n=1 \quad k \leq \frac{2D}{\lambda} = 141.2 \quad k_{\max} = 141$$

共有 142 条暗条纹.



二. 劈尖干涉的应用.

1. 干涉膨胀仪.



$$2d + \frac{\lambda}{2} = k\lambda$$

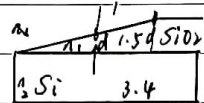
d 随膨胀, 条纹移动

$$2d_2 + \frac{\lambda}{2} = k_2\lambda \quad \text{条纹移动的数目}$$

$$2d_1 = N\lambda$$

$$d_1 = \frac{1}{2}N\lambda = N\frac{\lambda}{2}$$

2. 测膜厚



$$\Delta r = 2n_1d + \frac{\lambda}{2} = k\lambda \quad k=0, 1, 2, \dots$$

$$2n_1d = (N-1)\lambda$$

$$d = \frac{(N-1)\lambda}{2n_1}$$

d=0.

明纹.

0级 ... N-1级

资料完全免费. 如付费购买, 请保留证据并联系: gubaibai.ovo@gmail.com

$$\Delta l = \frac{\lambda}{2n\theta}$$

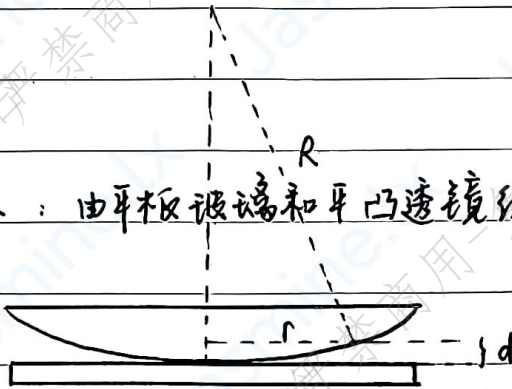
02.

3. 光学元件表面

4.

二. 牛顿环：由平板玻璃和平凸透镜组成。

空气的上下侧表面



$$\Delta l = \frac{\lambda}{2n\theta}$$

光程差 $\delta = 2nd + \frac{\lambda}{2} = 2d + \frac{\lambda}{2}$

(间距不均).

干涉图样：内疏外密，环状结构，中心暗斑(反射光)。

$$\delta = 2d + \frac{\lambda}{2} = \begin{cases} k \cdot \lambda & k=1, 2, \dots \\ (2k+1) \cdot \frac{\lambda}{2} & k=0, 1, 2, \dots \end{cases}$$

$$(R-d)^2 + r^2 = R^2$$

$$r^2 = 2Rd - d^2 \approx 2Rd + o(d)$$

$$r^2 = 2Rd$$

$$r = \sqrt{2dR} = \sqrt{(2d - \frac{\lambda}{2})R}$$

明环： $r = \sqrt{(k - \frac{1}{2})R\lambda} \quad (k=1, 2, \dots)$

$r = \sqrt{kR\lambda} \quad (k=0, 1, 2, \dots)$

单缝 光栅

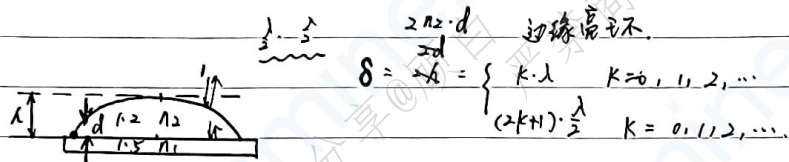
透镜曲率半径

$$r_k^2 = kR\lambda$$

$$r_{k+m}^2 = (k+m)R\lambda$$

$$\Rightarrow R = \frac{r_{k+m}^2 - r_k^2}{m\lambda}$$

e.g. 4



$$2n_2 d = k\lambda \quad (k=0, 1, 2, \dots)$$

$$d = \frac{k\lambda}{2n_2} = \frac{600 \times k}{2 \times 1.2} = 250k \quad (d \leq 1.5 \text{ mm})$$

$k=0 \quad d=0$

$k=1 \quad d=250 \text{ nm}$

看到4条明纹环状分布

$k=2 \quad d=500 \text{ nm}$

$k=3 \quad d=750 \text{ nm}$

§11-6 光的衍射

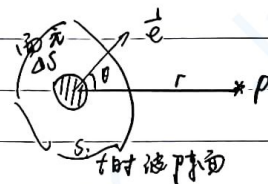
一. 光的衍射现象 (diffraction).

二. 光源、缝、屏相距有限远 菲涅耳衍射 (近场).

光源、缝、屏相距无限远 夫琅禾费衍射 (远场)

三. Fresnel 原理

波场中各点的强度由各子波在该点的相干叠加决定.

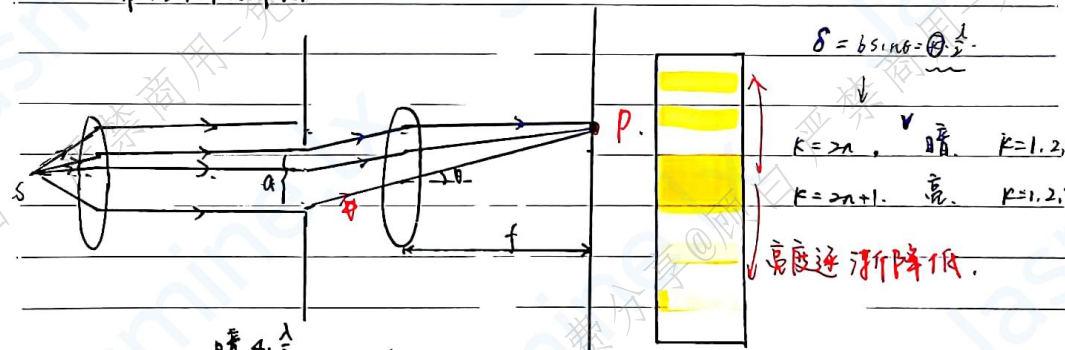


θ : 衍射角

§11-7 单缝衍射

夫琅禾费单缝衍射.

菲涅尔半波带法.



$$K=2 \quad b \sin \theta = 2 \cdot \frac{\lambda}{2} = \lambda$$

一个半波带

$$\frac{1}{2}b = 3\frac{\lambda}{2}$$

$$b \sin \theta = 2 \cdot \frac{\lambda}{2} = \lambda$$

$$b = 2\lambda$$

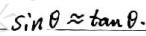
$$b \sin \theta = \frac{\lambda}{2}$$

$k=1$ 暗

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随着级数的增大, 角就增大, $\sin t$ 增大 分成的半波带越多

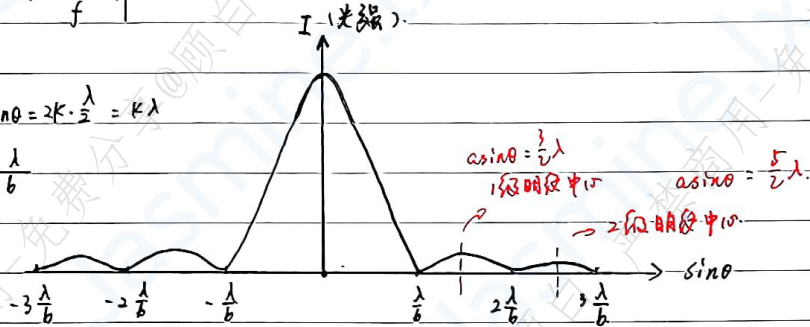
二. 光强分布: 中间最强, 逐级递减 本级比本级中更强 $\frac{I_1}{I_0} = \sin^2 \frac{\alpha}{2}$
心光强越_小, 亮度越_暗



$$x = f \tan \theta = f \sin \theta.$$

暗纹 $b \sin \theta = 2k \cdot \frac{\lambda}{2} = k\lambda$

$$\sin \theta = K \frac{\lambda}{b}$$


$$\cos \theta = \frac{1}{2}$$
$$a \sin \theta = \frac{5}{2} \lambda$$
$$-3\frac{\lambda}{b}f \quad -2\frac{\lambda}{b}f \quad -\frac{\lambda}{b}f \quad \frac{\lambda}{b}f \quad 2\frac{\lambda}{b}f \quad 3\frac{\lambda}{b}f.$$
$$\begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ k=-3 & k=-2 & \sin \theta, k=1 \end{array}$$

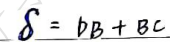
↑ ↑ ↑
 $k=1$ $k=2$ $k=3$

中央明纹宽度: $l_0 = 2 \frac{\lambda}{b} f$

$\lambda \downarrow$ b 改变 $b \uparrow$ $\omega \downarrow$ $\frac{\lambda}{b} \rightarrow 0$ 光沿直线传播

★ 单缝衍射的初态变化。 → 抓住：透镜中心的光线路径是否改变

单缝上下移, 衍射图像不变; 透镜 $\uparrow$ 移, ... 图像改变



$$= b(\sin \theta + \sin \varphi).$$

$$\delta = 0 \quad \theta = -\varphi$$

中央明纹向下移动。



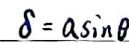
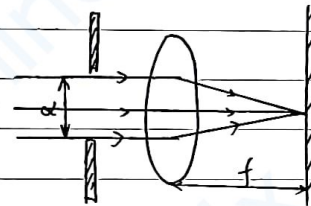
外:

中央明纹向右移动

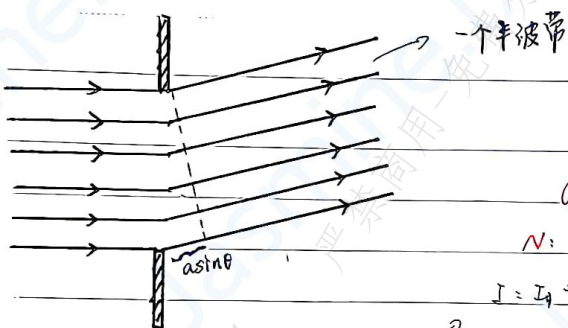
菲涅尔半波带法.

中央明纹中: $\theta=0$ $\delta=0$

中央明纹光强最大, 光能绝大部分集中在中央明纹内



将缝处波阵面分成若干等份, 使每个等份相对应的光线的光程差为 $\frac{\lambda}{2}$.



$$a \sin \theta = N \cdot \frac{\lambda}{2}$$

N : 半波带数 $\rightarrow$ 由 $a \sin \theta$ 决定

$$I = I_0 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$= 2I_0 + 2I_0 \cos 2\pi \cdot \frac{1}{\lambda} \cdot 0 \rightarrow \text{暗纹}$$

1° $a \sin \theta = \lambda$ $N=2$, 可将缝分为两个半波带.

两个半波带发出的光在 P 处 相干相消 形成暗纹中心 (一级暗纹).

2° $a \sin \theta = \frac{3}{2}\lambda$ $N=3$ 可将缝分为三个半波带

将两个半波带发出的光在 P 处 叠加, 形成明纹中心 (一级明纹).

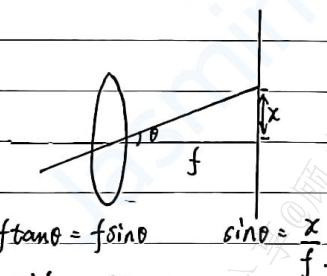
:

$$\delta = a \sin \theta = \begin{cases} \pm k\lambda & k=1, 2, \dots \text{暗纹中心} \\ \pm (2k+1)\frac{\lambda}{2} & k=1, 2, \dots \text{明纹中心} \\ 0 & \text{中央明纹中心} \end{cases}$$

中央明纹中心 — 衍射主极大

明纹中心 — 衍射次极大

暗纹中心 — 衍射极小



条纹宽度分析

暗纹中心: $a \sin \theta = a \frac{x}{f} = \pm k\lambda$

$$x = f \tan \theta \approx f \sin \theta \quad \sin \theta = \frac{x}{f}$$

$$x_k = \pm \frac{k\lambda f}{a} \quad (k=1, 2, \dots)$$

明纹中心: $a \sin \theta = a \frac{x}{f} = \pm (2k+1) \frac{\lambda}{2}$ $x_k = \pm \frac{(2k+1) \cdot \lambda f}{2a} \quad (k=1, 2, \dots)$

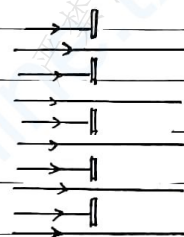
1° 中央明纹宽度: $\Delta x_0 = 2\lambda f / a$

2° 其他级次明纹宽度: $\Delta x = \frac{\lambda f}{a}$

($\theta < 5^\circ$ 上述近似成立!)

衍射光栅.

1. 光栅: 大量等宽等间距的平行狭缝组成的光学器件.



2. 光栅常量



a : 透光部分, 即缝宽.

b : 不透光的部分

d : 光栅常量, 两缝中心距离.

$$a+b=d \text{ (光栅常量)}$$

光栅宽度 = $N \times d$
总缝数

eg. 每毫米有 800 条刻度的光栅, 光栅常量为?

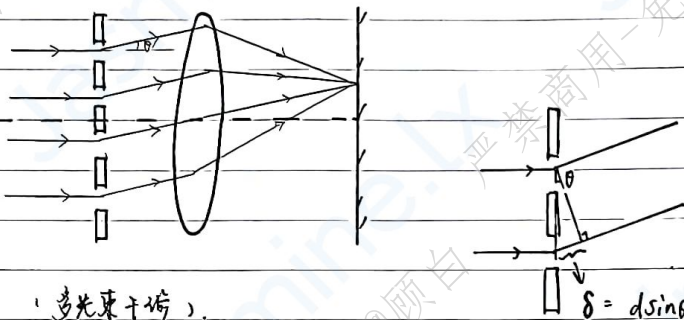
宽度 $\downarrow$ 缝数 $\downarrow$

$$d = \frac{1}{800} = 1.25 \times 10^{-3} \text{ mm}$$

$$N \cdot d \sin \theta' = \lambda$$

$$\theta' = \frac{\lambda}{Nd}$$

3. 光栅衍射



讨论干涉 (多光束干涉)

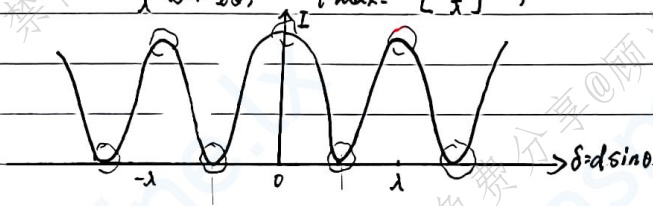
当 $d \sin \theta = \pm k \cdot \lambda$ $k=0, 1, 2, \dots$ 时相干相长,

形成明条纹 (即主极大条纹)

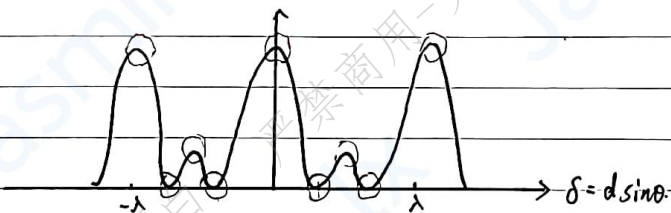
$$k_{\max} = \frac{d}{\lambda} \quad \text{且 } \frac{d}{\lambda} = m \in \mathbb{Z}, \quad k_{\max} = \frac{d}{\lambda} - 1;$$

$\frac{d}{\lambda}$ 为小数, $k_{\max} = [\frac{d}{\lambda}]$

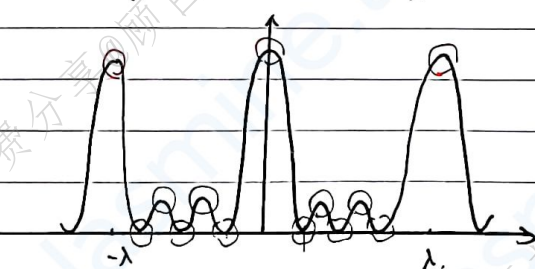
$N=2$



$N=3$



$N=4$



$$\delta = \frac{\lambda}{2}$$

(2) 孔衍射

规律: ① $\delta = \pm k \cdot \lambda$ 时为第 k 级主极大, 主极大为又窄又亮的明纹

② 相邻主极大间有 $N-1$ 个极小, $N-2$ 个次极大

③ 狭缝条数越多, 明纹越细; 次极大与极小混在一起, 形成暗区

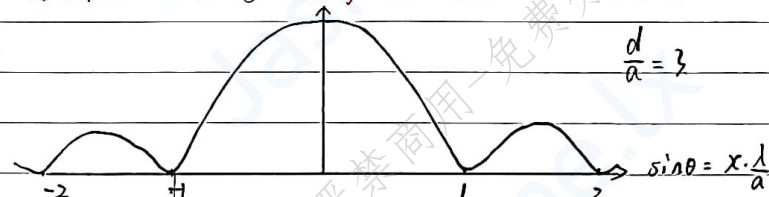
主极大明条纹变得很窄

单缝衍射 + 多光束干涉 $\Rightarrow$ 光栅衍射

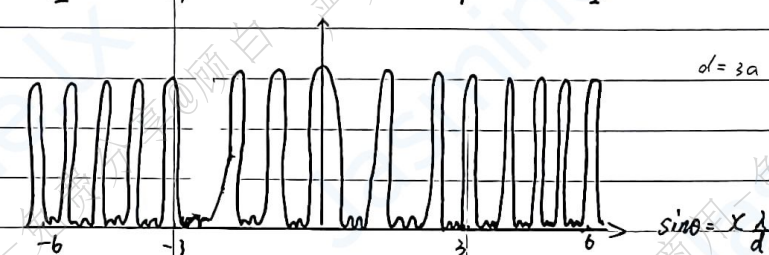
缺级

若主极大: 位置由多光束干涉决定, 强度受单缝衍射影响

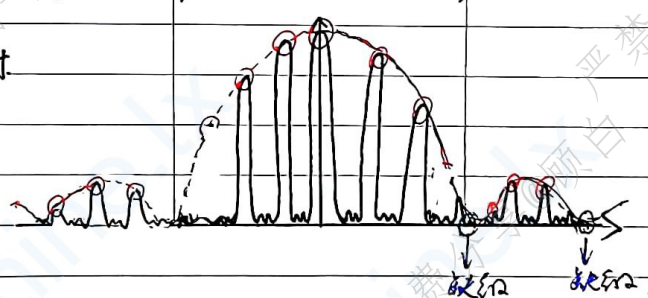
单缝衍射



多光束干涉



光栅衍射



$$a = a \quad b = a \quad d = 3a$$

二. 光栅方程:

相邻两缝光线的相位差为 $(a+b)\sin\theta$

光栅方程为 $(a+b)\sin\theta = \pm k\lambda \quad k=0, 1, 2, \dots$

对应主极大明纹

屏上明条纹位置 $x = f \tan\theta \quad (\theta \neq \sin\theta \neq \tan\theta)$

主极大级明纹

三. 缺级现象:

多光束干涉相长条件: $(a+b)\sin\theta = \pm k\lambda \quad k=0, 1, 2, \dots$
 单缝衍射暗纹条件: $a\sin\theta = \pm k'\lambda \quad k'=1, 2, 3, \dots$

当同时满足上述两方程, 则光栅衍射中 k 级主极大消失

主极大缺级

$k = \frac{a+b}{a} k'$
 k 只能取整数

衍射暗纹级次

缺级现象

三. 衍射光谱: (读教材)

d 光栅

$(b+b')\sin\theta = 3\lambda_{紫}$
 $(b+b')\sin\theta = 2\lambda$

中央主极大位置

$k = \frac{a}{b} k' \quad k \neq 0$
 $d = 7b$

$d\sin\theta = k\lambda$

光谱分析:

用白光 (400nm ~ 760nm) 垂直照射每厘米有 4000 条缝的光栅, 问产生多少级完整清晰可见的光谱: 第 2 级光谱与第 3 级光谱是否重叠?

$d = 1\text{cm} / 4000 = 2.5 \times 10^{-4} \text{cm}$

由光栅方程 $d\sin\theta = k\lambda$

$k_{\max} \quad \theta = \frac{\pi}{2} \quad k = \frac{d}{\lambda} \quad \lambda = \lambda_{红} \quad k = 2.5 \times 10^{-6} / 0.76 \times 10^{-6} = 3.29$

$\therefore$ 可产生 3 级完整清晰可见的光谱.

$\begin{cases} d\sin\theta = 3 \cdot \lambda_{紫} \\ d\sin\theta = 2 \cdot \lambda \end{cases} \quad \lambda = \frac{3}{2} \lambda_{紫} = \frac{3}{2} \times 400 = 600 \text{nm} \in (400\text{nm}, 760\text{nm})$

$\therefore$ 2, 3 两级光谱有重叠.

衍射光谱

光谱级次

1 级明纹

入射光为白光时, 形成彩色光谱

①

同级不同颜色的明条纹按波长顺序排列, 形成衍射光谱

2 级光谱重叠光谱范围:

$(b+b')\sin\theta = 3\lambda_{紫}$
 $(b+b')\sin\theta = 2\lambda \quad \lambda = \frac{3}{2} \lambda_{紫} = 600 \text{nm}$

同处
 θ 处

2 级光谱重叠 $600\text{nm} \sim 760\text{nm}$

部分:

e.g. 用白光垂直照射在每厘米 6500 条刻痕的平面光栅上, 求第 3 级光谱的级角.

光栅常数 $d = 1\text{cm} / 6500 = 1.54 \times 10^{-6} \text{m}$

$d\sin\theta = k\lambda$

$\sin\theta = \frac{k\lambda}{d} = \frac{3}{1.54 \times 10^{-6}} \cdot \lambda$

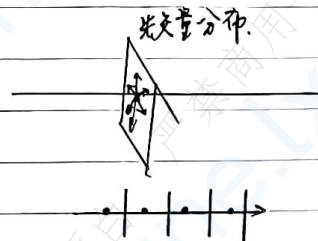
$\lambda = 400 \text{nm} \quad \sin\theta = \frac{3 \times 400 \times 10^{-9}}{1.54 \times 10^{-6}} = 0.7792 \quad \theta = 51.19^\circ$

本资料完全免费。如遇付费购买, 请保留证据并联系: gubaibai.ovo@foxmail.com

$\lambda = 760 \text{nm} \quad \sin\theta = \frac{3 \times 760 \times 10^{-9}}{1.54 \times 10^{-6}} > 1 \quad \therefore \theta \in [51.19^\circ, 90^\circ]$

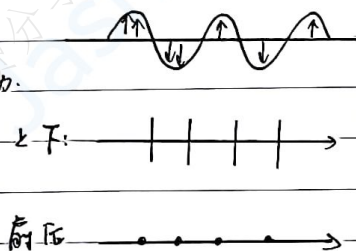
§ 11-10 光的偏振.

1. 自然光



2. 线偏振光

只沿某一方向振动.



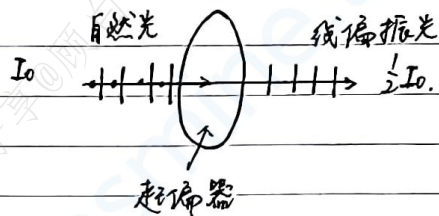
3. 部分偏振光.

二. 偏振片 起偏与检偏

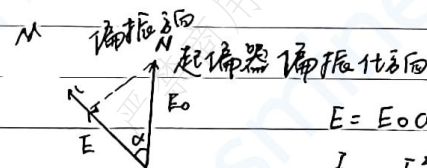
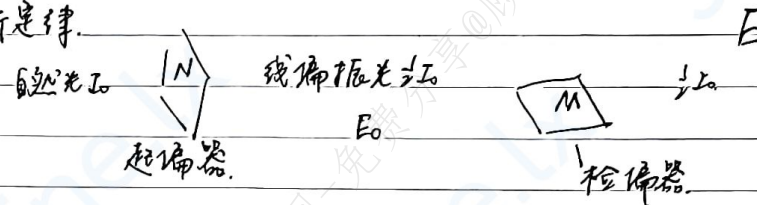
1. 二向色性

2. 偏振片

3. 偏振化方向



三. 马吕斯定律.

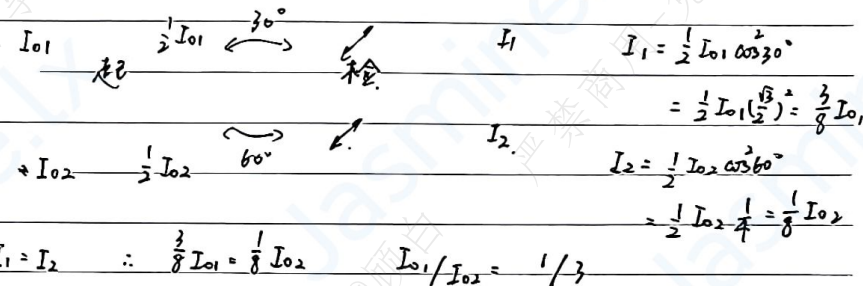


$$E = E_0 \cos \alpha$$

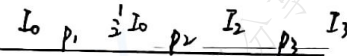
$$\frac{I}{I_0} = \frac{E^2}{E_0^2} = \cos^2 \alpha$$

$$\therefore I = I_0 \cos^2 \alpha$$

e.g. 1



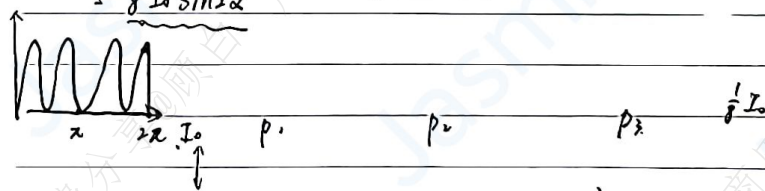
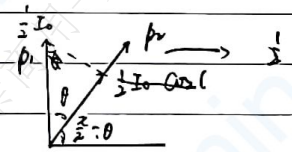
e.g. 2



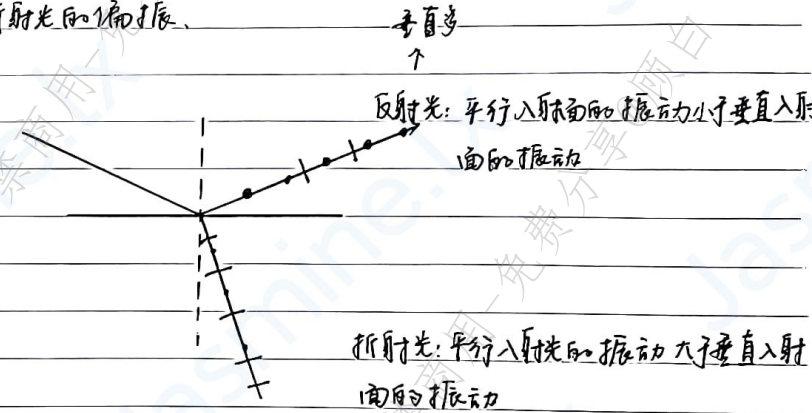
$$I_1 = \frac{1}{2} I_0$$

$$I_2 = \frac{1}{2} I_0 \cos^2 \alpha$$

$$I_3 = \frac{1}{2} I_0 \cos^2 \alpha \sin^2 \alpha = \frac{1}{8} I_0 \sin^2 2\alpha$$



§ 11-11 反射光、折射光的偏振

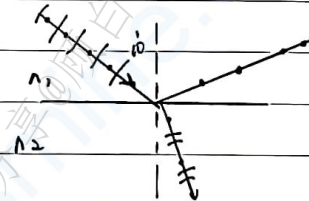


2. Brewster's law.

当 $\tan i_0 = \frac{n_2}{n_1}$ 时,
反射光为完全偏振光,
且振动面垂直于入射面.
折射光为部分偏振光.

(由光路可逆性, i_0 亦称为布儒斯特角).

$n_2 > n_1$ 自然光



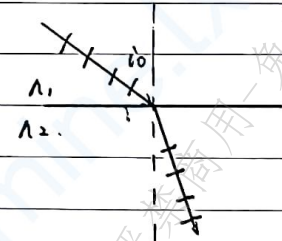
布儒斯特角起偏角

$$\frac{\sin i_0}{\sin r_0} = \frac{n_2}{n_1} \quad \text{即特进入} \quad \text{光}$$

$$\tan i_0 = \frac{n_2}{n_1} = \frac{\sin i_0}{\cos i_0}$$

$$n_2 \therefore \sin r_0 = \cos i_0$$

$$\therefore i_0 + r_0 = \frac{\pi}{2} \quad \text{光折射率无失}$$



第12章 气体动理论

12-1 平衡态 理想气体状态方程

宏观量 大量分子 p, V, T 可直接测量

微观量 个别分子 $m, \vec{v}$ 不可直接测量

微观量 $\xrightarrow{\text{统计平均}}$ 宏观量

一. 气体的状态参量

$$1 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$$

1. 压强 p : 力学描述 (Pa)

2. 体积 V : 几何描述 $1 \text{ m}^3 = 10^3 \text{ L}$

3. 温度 T : 热学描述 K $T = 273.15 + t$

二. 平衡态

三. 理想气体状态方程

一定量的气体, 温度不太低, 压强不太大

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

物质的量

系统总质量

$$pV = \nu RT$$

$$\frac{pV}{T} = \nu \cdot R$$

摩尔气体常数

$$R = 8.31 \text{ J/(mol} \cdot \text{K)}$$

$$\frac{m'}{m} \cdot R$$

摩尔质量

$$p = n \cdot k T$$

$$\frac{N}{V}$$

数密度

$$pV = \nu RT = \frac{m'}{m} RT = \frac{m \cdot N}{m \cdot N_A} RT =$$

$$p = \frac{N}{V} \frac{R}{N_A} T = n \cdot k T$$

$$N_A = 6.02 \times 10^{23}$$

气体分子数密度

$$k = \frac{R}{N_A}$$

玻耳兹曼常数

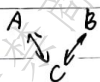
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$$k = 1.38 \times 10^{-23}$$

45 90%

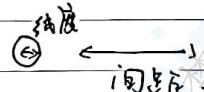
四. 热力学第零定律

1. 热平衡:
2. 热力学第零定律.

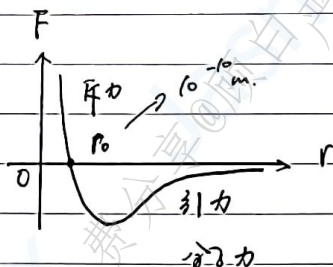


§12-2. 物质的微观模型 统计规律性.

一. 分子的线度和分子力



$r < r_0$ 斥力
 $r > r_0$ 引力



二. 分子热运动的无序性及统计规律

$$w_i = \lim_{N \rightarrow \infty} \frac{N_i}{N} \quad \text{归一化条件: } \sum_i w_i = \sum_i \frac{N_i}{N} = 1$$

§12-3. 理想气体的压强公式.

一. 理想气体的微观模型.

热动平衡下的统计规律 (平衡态):

1. 每个分子的运动速度各不相同.
2. 分子按位置的分布是均匀的.
3. 分子各方向运动概率相等

$$\bar{v_x} = \bar{v_y} = \bar{v_z} = \frac{1}{3} \bar{v} \\ \bar{v_x} = \bar{v_y} = \bar{v_z} = 0$$

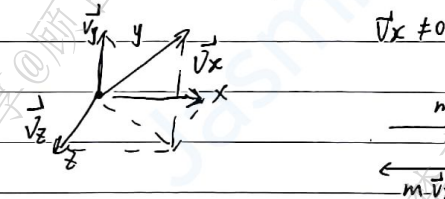
二. 理想气体压强公式.

单个分子碰撞 - 偶然性 大量分子 恒定的持续力作用
不连续性



N 个全同的质量为 m 的分子.

求 A1 所受压强.



x 方向的动量变化: $\Delta p_{ix} = -2m v_{ix}$

分子对器壁的冲量: $2m v_{ix}$

两次碰撞时间间隔: $\frac{2x}{v_{ix}}$

$$F_{Ax} = m \cdot a_x$$

单位时间碰撞次数: $v_{ix}/2x$

单个分子单位时间

的冲量: $2m v_{ix} \cdot \frac{v_{ix}}{2x}$

$$\frac{m v_{ix}^2}{x}$$

单位时间总冲量 $\rightarrow$ 冲力

$$= \frac{m v_{ix}^2}{x}$$

$$m \cdot \Delta v$$

$$大量的: \bar{F} = \sum_i \frac{m v_{ix}^2}{x} = \frac{Nm}{x} \sum_i \frac{v_{ix}^2}{N} = \frac{Nm}{x} \bar{v_x^2}$$

$$\text{冲力 } \bar{F} = \frac{1}{3} N m \bar{v^2}$$

$$p = \frac{\bar{F}}{y z} = \frac{Nm}{x y z} \bar{v_x^2} = \frac{1}{3} n m \bar{v^2}$$

分子平均动能: $\bar{\epsilon_k} = \frac{1}{2} m \bar{v^2}$

统计规律: $\bar{v_x^2} = \frac{1}{3} \bar{v^2}$

平均力

$\downarrow$

总平均动能

$$p = \frac{2}{3} n \bar{\epsilon_k} = \frac{2}{3} \frac{N \bar{\epsilon_k}}{V}$$

数密度 平均动能

$\rightarrow$ 体积

§12-4. 理想气体的平均平动 $\bar{\epsilon}_k \propto T$ $n = \frac{N}{V}$

初能:

$$p = \frac{2}{3} n \bar{\epsilon}_k \quad p = n k T$$

↓

$$\bar{\epsilon}_k = \frac{1}{2} m \bar{v}^2 = \frac{3}{2} k T \quad \bar{\epsilon}_k = \frac{3}{2} k T$$

↑
平均动能

↑
玻尔兹曼常数

微观量的统计平均

1. 温度是分子平均平动动能的量度 $\epsilon_k \propto T$

2. 温度是大量分子运动的集体表现

3. $T \propto \epsilon_k$ 与物体种类无关

§12-5. 能量均分定理

- 自由度: 确定一个物体空间位置所需要的独立坐标数

单原子分子 3

刚性双原子分子: $2 \times 3 - 1 = 5$

$\frac{5}{2}$

直线: 2

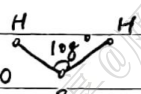
转动: 3

$3 + 2 = 5$

刚性多原子分子

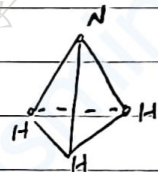
$$3 \times 3 - 1 \times 3 = 6$$

$n = 3$ H₂O



$n = 4$

NH₃



$$4 \times 3 - 6 = 12 - 6 = 6$$

n

$n = 5$

CH₄



$$n = 3 - \frac{C_n^2}{2}$$

$$5 \times 3 - 6 = 3 = 6$$

★

刚性分子能量自由度

	平动自由度	转动自由度	总
单原子分子	3	0	3
双原子分子	3	2	5
多原子分子	3	3	6

1. 自由度

能量均分定理: 平衡态, 的任何一个自由度的平均能量都相等, 均为 $\frac{1}{2} k T$

$$\bar{\epsilon}_k = \frac{1}{2} m \bar{v}^2 = \frac{1}{2} m \bar{v}_x^2 + \frac{1}{2} m \bar{v}_y^2 + \frac{1}{2} m \bar{v}_z^2 \quad \text{平均平动 } \bar{\epsilon}_k = \frac{3}{2} k T$$

$$\frac{1}{2} m \bar{v}_x^2 = \frac{1}{2} m \bar{v}_y^2 = \frac{1}{2} m \bar{v}_z^2 = \frac{1}{2} k T$$

平动自由度都为3
(无论何种原子)

分子的平均能量: $\bar{\epsilon} = \frac{1}{2} k T$

O₂ 平均平动动能

eg. 1. 求温度为 20°C 时氧分子的平均能量 $\bar{\epsilon}$ (平均能量) $\frac{1}{2} k T$

$$\bar{\epsilon}_k = \frac{3}{2} k T = \frac{3}{2} \times 1.38 \times 10^{-23} \times 293 = 6.07 \times 10^{-21} \text{ J} = 3.8 \times 10^{-2} \text{ eV}$$

双原子分子

$$(O_2) \quad 2 \times 3 - 1 = 5$$

$$1 \text{ J} = 6.25 \times 10^{18} \text{ eV}$$

$$\bar{\epsilon} = \frac{5}{2} k T = 63 \text{ meV}$$

eg. 2. 某原子以恒定速率 v 沿 z 方向做整体运动, 求 $\bar{\epsilon}_k$

$$\frac{1}{2} m v^2 = \frac{1}{2} k T$$

$$\bar{\epsilon} = \frac{3}{2} k T + \frac{1}{2} m v^2 = \frac{3}{2} k T + \frac{1}{2} k T = 2 k T$$

$$\bar{\epsilon} = \bar{\epsilon}_1 + \bar{\epsilon}_2 = \frac{3}{2} k T + \frac{1}{2} m v^2$$

无规则
热运动

整体

3. 理想气体的内能: 分子动能和分子势能之和

理想气体不考虑势能.

1 mol 理想气体, $E = 1 \cdot N_A \cdot \bar{\epsilon} = \frac{i}{2} RT$

$\bar{\epsilon} = \frac{R}{N_A} T$

ν mol 理想气体的内能: $E = \nu \cdot \frac{i}{2} RT$

$R = \frac{R}{N_A}$

$E = E(T)$

状态 $V \rightarrow T \rightarrow E$

状态量.

$dE = \nu \cdot \frac{i}{2} R \cdot dT$

A. 1 mol O₂ B. 1 mol CH₄

内能相同.

平均转动动能之比

$E = \nu \cdot \frac{i}{2} RT$

$\bar{\epsilon}_R = \frac{i}{2} \frac{R}{N_A} T$

$3 \times 2 = 6$

$E_A = E_B$

$\frac{5}{2} R T_1 = \frac{6}{2} R T_2$

$\frac{T_1}{T_2} = \frac{6}{5}$

$\frac{\bar{\epsilon}_{T1}}{\bar{\epsilon}_{T2}} = \frac{\frac{2}{2} K T_1}{\frac{3}{2} K T_2} = \frac{2}{3} \times \frac{6}{5} = 4:5$

$E_A = E_B$

$\frac{5}{2} R T_1 = \frac{6}{2} R T_2$

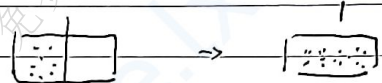
$\frac{T_1}{T_2} = \frac{6}{5}$

$\frac{\bar{\epsilon}_{T1}}{\bar{\epsilon}_{T2}} = \frac{\frac{2}{2} K T_1}{\frac{3}{2} K T_2} = \frac{2}{3} \times \frac{6}{5} = 4:5$

热力学基础. $\rightarrow$ 热力学第一定律.

准静态过程

绝热自由膨胀 非准静态过程.

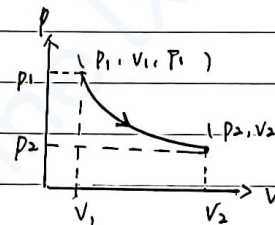


热力学第一定律

热量 内能 过程量

本质是分子间的碰撞.

准静态过程功的计算.



$dw = F dl = p S dl = p \cdot dV$
 $dw = p dV$

膨胀 $dV > 0$ $dw > 0$

$V_2 > V_1$

$dV < 0$ $dw < 0$ 气体对外做负功.

外对气体做正功.

$w = \int_{V_1}^{V_2} p dV$ 功与过程有关.

仅知道初末态无法确定 w .

e.g. ν mol 气体 T 不变 V_1 准静态 V_2 , 系统对外做功.

$dw = p dV$

$w = \int dw = \int_{V_1}^{V_2} p dV = \nu R T \int_{V_1}^{V_2} \frac{1}{V} dV$

理想气体 $pV = \nu RT$

$= \nu R T \ln \frac{V_2}{V_1}$

初态 V_1 终态 V_2 气体膨胀: 对外做功

$V_2 < V_1$ 气体压缩 外对气体做功.

热量

$$T_1 < T_2 \quad T_1 \uparrow \uparrow Q$$

功、热量

1. 都是过程量，与过程有关。

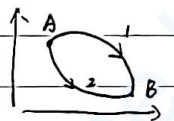
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内能

$$E = E(T) = \frac{1}{2} \nu R T$$



$$\Delta E_{AB} = C$$



$$\Delta E_{AB} > A = 0$$

热力学第一定律

系统从外界吸收的热量，一部分使系统的内能增加，另一部分使系统对外界作功

$$Q = E_2 - E_1 + W = \Delta E + W$$

准静态过程

$$Q = \Delta E + \int_{V_1}^{V_2} p dV$$

$$dQ = dE + dw$$

$$= dE + p \cdot dV$$

$$Q = \Delta E + W$$

系统从外界吸收能量 $\uparrow$ 系统内能增加 $\downarrow$ 系统对外作功 $\downarrow$

系统吸热 + $V \uparrow$ +
系统放热 - $V \downarrow$ -

$T \uparrow$ + $T \downarrow$ -

1. 能量转换和守恒定律 $\rightarrow$ 第一类永动机是不可能制成的。

热力学第一定律的应用

$$1. PV = \nu R T \quad (\text{理想气体状态方程})$$

$$2. \begin{cases} dQ = dE + p dV \\ Q = E + \int_{V_1}^{V_2} p dV \end{cases}$$

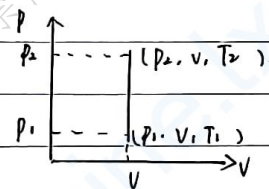
$$3. E = E(T) \quad (\text{内能是状态量})$$

绝热过程 $Q = 0$

1. 气体 (各过程)

$$PV = \nu R T$$

$$p \propto T$$



$$W = \int_{V_1}^{V_2} p dV \quad dV = 0 \quad \therefore W = 0$$

$$dQ = dw + dE \quad \therefore dQ = dE$$

系统从外界吸收能量 = 系统内能

摩尔定体热容

1 mol 理想气体在定体过程中

$$C_{v,m} = \frac{dQ_v}{dT} \quad dQ_v = C_{v,m} dT$$

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摩尔定体热容

热 -
 V 恒定: 对 V mol 气体
 $dQ_V = dE = \nu C_{V,m} dT$

$$dE = \nu \cdot \frac{i}{2} R dT \quad C_{V,m} = \frac{i}{2} R$$

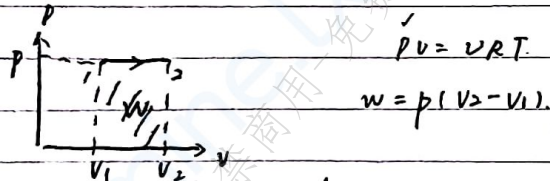
↑
摩尔气体常数

$Q_V = \Delta E$
 气体: $Q_V = \nu C_{V,m} \Delta T = E_2 - E_1$

$$dE = \nu C_{V,m} dT \quad C_{V,m} = \frac{i}{2} R$$

气体 / 气体膨胀
 $\uparrow pV = \nu R T$
 $E \uparrow \Rightarrow$ 系统从外界吸热

定压过程



$$dQ_p = dE + dW$$

↓
系统对外做功

摩尔定压热容, 1 mol 气体 ...

$$C_{p,m} \frac{dQ_p}{dT} \quad dQ_p = C_{p,m} dT$$

考虑 1 mol 气体 $pV = RT \rightarrow p dV = R dT$

$$dQ_p = dE + dW$$

$$C_{p,m} dT = C_{V,m} dT + p dV = C_{V,m} dT + R dT$$

迈耶公式: $C_{p,m} = C_{V,m} + R = (\frac{i+2}{2}) \cdot R$

摩尔热容比: $\gamma = \frac{C_{p,m}}{C_{V,m}} = \frac{i+2}{i}$

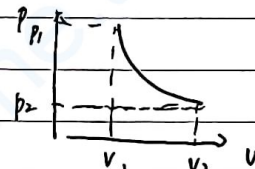
单原子分子: $i=3, C_{V,m} = \frac{3}{2} R, C_{p,m} = \frac{5}{2} R, \gamma = \frac{5}{3}$

膨胀
 收缩

单位质量的热容

热容 $C = dQ/dT$ 比热容 $c = \frac{dQ}{m dT} = \frac{C}{m}$

升温过程



$$pV = \nu R T \quad pV = \nu R T$$

$$dQ_T = dE + dW \quad dW = \int_{V_1}^{V_2} p dV = 0 + dW \quad W = \nu R T \ln \frac{V_2}{V_1}$$

气体对外

$$Q = \Delta E + W \quad W = \int_{V_1}^{V_2} p dV$$

$$dQ = dE + dW = dE + p dV$$

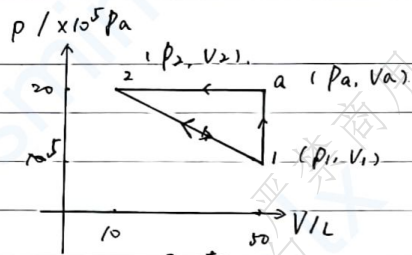
气体 (等压) $W=0 \quad Q = \Delta E = \nu \frac{i}{2} R \Delta T = \nu C_{V,m} \Delta T \quad C_{V,m} = \frac{i}{2} R$
 摩尔定压热容

定压 $pV \quad Q_p = \nu C_{p,m} \Delta T$
 $C_{p,m} = \frac{i+2}{2} \cdot R$ 摩尔定压

升温

$$\Delta E = \nu R T \ln \frac{V_2}{V_1}$$

e.g.



$\nu = 20 \text{ mol } O_2$

1. $1 \rightarrow a \rightarrow 2$

2. $1 \rightarrow 2$

1. $1 \rightarrow a \rightarrow 2$ 过程:

对 $1 \rightarrow a$: 等体过程: $W_{1a} = 0$

$$Q_{1a} = \Delta E_{1a} = \nu \cdot C_{v,m} \Delta T = \nu \cdot C_{v,m} (T_a - T_1)$$

$$= \nu \cdot \frac{i}{2} R (T_a - T_1) = \frac{i}{2} (\nu R T_a - \nu R T_1)$$

$$pV = \nu R T$$

$$= \frac{i}{2} (p_a V_a - p_1 V_1)$$

$$= \frac{1}{2} \times 5 \times 10^5 \times 50 \times 10^{-3}$$

$$= 1.875 \times 10^5 \text{ J}$$

对 $a \rightarrow 2$: 等压过程: $W_{a2} = p_a (V_2 - V_a)$

$$= 20 \times 10^5 \times (-40) \times 10^{-3} = -0.8 \times 10^5 \text{ J}$$

$$Q_{a2} = \nu \cdot C_{p,m} (T_2 - T_a)$$

$$= \nu \cdot \frac{i+2}{2} R (T_2 - T_a) = \frac{i+2}{2} (\nu R T_2 - \nu R T_a)$$

$$Q_{a2} = \frac{i+2}{2} (p_2 V_2 - p_a V_a) = \frac{1}{2} \times 7 \times 20 \times 10^5 \times (-40) \times 10^{-3} = -2.8 \times 10^5 \text{ J}$$

$$Q_{a2} = \nu \cdot C_{v,m} (T_2 - T_a) = -2 \times 10^5 \text{ J}$$

$$\text{or } (\Delta E)_{a2} = Q_{a2} - W_{a2} = -2 \times 10^5 \text{ J}$$

$$\text{对 } 1 \rightarrow a \rightarrow 2: W_1 = W_{1a} + W_{a2} = -0.8 \times 10^5 \text{ J} \quad Q_1 = Q_{1a} + Q_{a2} = -0.925 \times 10^5 \text{ J}$$

$$\Delta E_1 = \Delta E_{1a} + \Delta E_{a2} = -0.125 \times 10^5 \text{ J}$$

ΔE

$V \downarrow \quad W < 0$

$$= 2.5 \times 10^5$$

$$= -125$$

$$(2). \quad 1 \rightarrow 2 \quad W_2 = \frac{p_1 + p_2}{2} (V_2 - V_1) = -0.5 \times 10^5 \text{ J}$$

$$\Delta E_2 = \nu \cdot C_{v,m} \Delta T = \nu \cdot \frac{i}{2} R (T_2 - T_1) = \frac{i}{2} (\nu R T_2 - \nu R T_1)$$

$$= \frac{i}{2} (p_2 V_2 - p_1 V_1) = \frac{5}{2} \times (200 - 250) \times 10^2 = -0.125 \times 10^5 \text{ J}$$

$$Q_2 = W_2 + \Delta E_2 = -0.625 \times 10^5 \text{ J}$$

内能是状态量, 与过程无关

功与热量是过程量.

二. 绝热过程: 与外界无热量交换的过程 (理想化过程).

$$dQ = 0$$

由热一律: $dW + dE = 0$

$$dW = -dE$$

$$dW > 0$$

$$dE < 0$$

消耗内能来对外做功

对外

$$dW < 0$$

$$dE > 0$$

外界对物体做功, 内能↑.

温度↑.

$$dW = -dE$$

$$dT = -\frac{p dV}{\nu C_{v,m}}$$

$$\begin{cases} p dV = -\nu C_{v,m} dT \\ pV = \nu RT \end{cases} \xrightarrow{\text{微分}} p dV + V dp = \nu R dT \quad \text{①}$$

$$p dV + V dp = -\nu R \frac{p dV}{\nu C_{v,m}}$$

$$p dV + V dp = -\nu R \frac{p dV}{\nu C_{v,m}}$$

$$p dW + V dp = -\frac{R}{C_{v,m}} p dV$$

$$(1 + \frac{R}{C_{v,m}}) p dV + V dp = 0$$

$$r p dV + V dp = 0$$

绝热过程方程

$$\frac{C_{v,m} + R}{C_{v,m}} = \frac{C_{p,m}}{C_{v,m}} = \gamma \quad \text{摩尔热容比}$$

$$\gamma = \frac{r}{\nu} \frac{dV}{dV} + \frac{dp}{p} = 0$$

$$pV^\gamma = C \quad pV = \nu R T$$

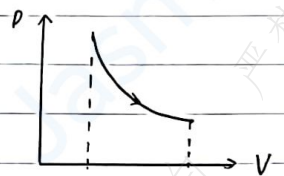
$$pV^\gamma = C \quad pV = \nu R T$$

$$\gamma = \frac{i+2}{i}$$

$$pV^\gamma = C \quad pV = \nu R T$$

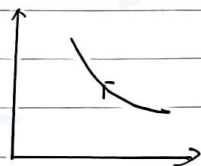
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绝热膨胀



内能↓ 部分
将内能转化为对外做功所耗能量

绝热压缩

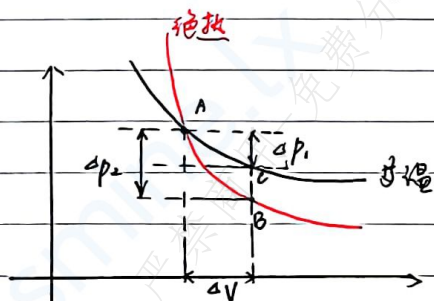


绝热: $pV^\gamma = C$

$$\gamma p V^{\gamma-1} dV + V^\gamma dp \quad \left(\frac{dp}{dV}\right)_a = -\gamma \frac{p}{V}$$

等温:

$$pV = C \quad pdV + Vdp = 0 \quad \frac{dp}{dV} = -\frac{p}{V}$$



绝热 等温

$$|\Delta p_2| > |\Delta p_1|$$

$$p = n \cdot k \cdot T$$

分子数密度

等温: $T \uparrow \quad p \downarrow \quad V \uparrow \quad n \downarrow$
 $V \uparrow \quad n \downarrow \Rightarrow p \downarrow$

绝热: $T \downarrow \quad n \downarrow \Rightarrow p \downarrow$

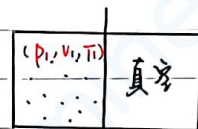
试分析绝热自由膨胀

非准静态过程

绝热过程方程

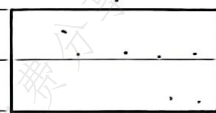
$$Q = \Delta E + W$$

$$Q = 0$$



向真空膨胀: $W = 0$

$$\Delta E = 0 \Rightarrow T_1 = T_2$$



$$(p_2, V_2, T_2)$$

$$\begin{cases} p_1 V_1 = \nu R T_1 \\ p_2 V_2 = \nu R T_2 \end{cases}$$

$$\Rightarrow p_1 V_1 = p_2 V_2$$

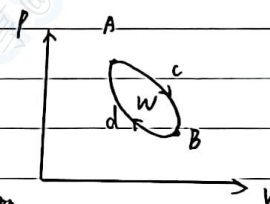
$$p_1 \cdot \frac{1}{2} V_2 = p_2 V_2 \quad \dots \quad p_2 = \frac{1}{2} p_1$$

§12-5 循环过程 卡诺循环

$$\Delta E = 0$$

$$\therefore Q = W$$

净功: $W = Q_1 - |Q_2| = Q$
 (总吸热) $\uparrow$ (总放热) $\downarrow$ (净吸热)



顺时针
(正循环)

$$W > 0$$

热机

从外界吸收热量 用于自身做功

逆时针

$$|Q_1| < |Q_2|$$

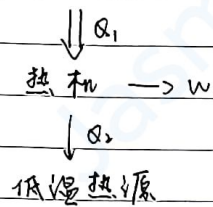
$$W < 0$$

制冷机

外界对物体做功

(逆循环)

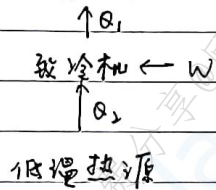
1. 高温热源



热机效率: $\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$

令取正

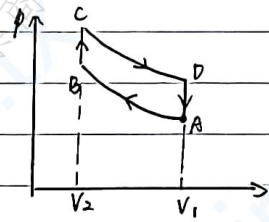
2. 高温热源



致冷系数: $e = \frac{Q_2}{|W|} = \frac{Q_2}{Q_1 - Q_2}$

$\frac{pV}{T} = C$

e.g. Otto 循环: AB, CD 为绝热过程 热机效率



A 吸入空气和汽油

AB 过程急剧压缩 → 绝热

B 电火花点火爆燃

BC 膨胀

C 推动活塞对外做功

CD

D 排出废气

AB: 绝热压缩 BC: 绝热膨胀 CD: 绝热膨胀 DA: 绝热压缩

$\eta = 1 - Q_2/Q_1$ 对 BC: $Q_{吸} = \nu C_{v,m} \Delta T = \nu C_{v,m} (T_C - T_B)$
对 BA: $Q_{放} = \nu C_{v,m} \Delta T = \nu C_{v,m} (T_D - T_A)$

$\therefore \eta = 1 - \frac{T_D - T_A}{T_C - T_B}$ 由绝热过程方程 $TV^{\gamma-1} = C$

$T_A V_A^{\gamma-1} = T_B V_B^{\gamma-1}$

$T_C V_C^{\gamma-1} = T_D V_D^{\gamma-1}$

$T_A/T_B = (V_2/V_1)^{\gamma-1}$

$T_D/T_C = (V_2/V_1)^{\gamma-1}$

$\therefore T_A/T_B = T_D/T_C = K$

$T_D = K T_C$

$T_D - T_A = K(T_C - T_B)$

$T_A = K T_B$

$\eta = 1 - K = 1 - \frac{T_A}{T_D} = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma-1}$

记 $r = \frac{V_1}{V_2}$ 压缩比

$\eta = 1 - \left(\frac{1}{r}\right)^{\gamma-1}$ 增函数 $r \uparrow \eta \uparrow$

一般取 $r = 7$ $r > 7$, T_B 超过混合气体燃点

$r = (1.4+2)/1.1 = 1.4$

$\Rightarrow \eta = 55\%$

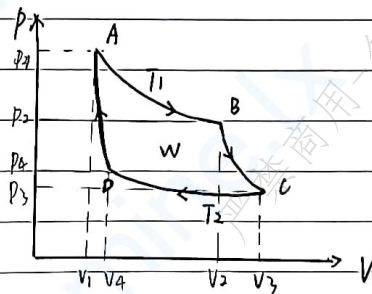
↓ 代表与外界有热交换

实际 25%

两个给定热源温度的条件下, 热机效率的理论极限值 $\eta = 1 - T_2/T_1$

三. 卡诺循环 Carnot cycle

由两个准静态等温过程和两个准静态绝热过程组成



AB 等温膨胀

CD 等温压缩并放热

$Q_1 = Q_{AB} = \nu R T_1 \ln \frac{V_2}{V_1}$

$Q_2 = |Q_{CD}| = \nu R T_2 \ln \frac{V_3}{V_4}$

$\eta = 1 - Q_2/Q_1 = 1 - \frac{T_2}{T_1} \ln \frac{V_3/V_4}{V_2/V_1}$

绝热过程方程 $TV^{\gamma-1} = C$

$T_1 V_2^{\gamma-1} = T_1 V_1^{\gamma-1}$

$T_2 V_3^{\gamma-1} = T_2 V_4^{\gamma-1}$

$\left(\frac{V_3}{V_4}\right)^{\gamma-1} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$

效率与工质性质无关, 只与两个热源的温差有关

$\eta = 1 - \frac{T_2}{T_1}$

温差越大, η 越大

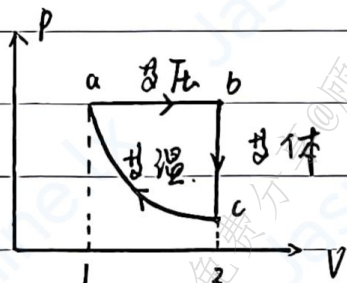
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e.g. 1 mol 单原子分子理想气体 ab 为等压线, bc 为等体线, ca 为等温线

$$T_c = 600 \text{ K}$$

(1) 各过程系统吸收的热量

(2) 循环效率



$$Q_{bc} = \Delta E = C_{V,m} \Delta T$$

$$T_a = 600 \text{ K}$$

$$T_a = T_c = 600 \text{ K} \quad T_b/T_a = V_b/V_a = 2 \quad \therefore T_b = 1200 \text{ K}$$

$$\text{吸 } Q_{ab} = \nu \cdot C_{p,m} (T_b - T_a) = \frac{5}{2} R \times 600 = 1.25 \times 10^4 \text{ J} > 0$$

$$\text{放 } Q_{bc} = \nu \cdot C_{V,m} (T_c - T_b) = \frac{3}{2} R \times (-600) = -0.75 \times 10^4 \text{ J} < 0$$

$$\text{放 } Q_{ca} = W_{ca} = \nu R T_a \ln \frac{V_a}{V_c} = -0.35 \times 10^4 \text{ J}$$

$$(2) \quad \eta = 1 - \frac{|Q_2|}{Q_1} = 1 - \frac{|Q_{bc}| + |Q_{ca}|}{Q_{ab}} = 10\%$$

§ 12-6 热力学第二定律

1. 功热转换

摩擦使功变热过程不可逆。

2. 热传导不可逆。

1. 开尔文表述

唯一效果是热全部转变成功的过程是不可逆的。

2. 克劳修斯表述

热量不能自动地从低温物体传向高温物体。

第十五章 量子物理

Date: _____

普朗克量子假设

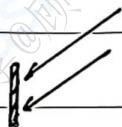
谐振子的能量状态是分立的

$$\varepsilon = n \cdot h\nu \quad (n=1, 2, 3, \dots)$$

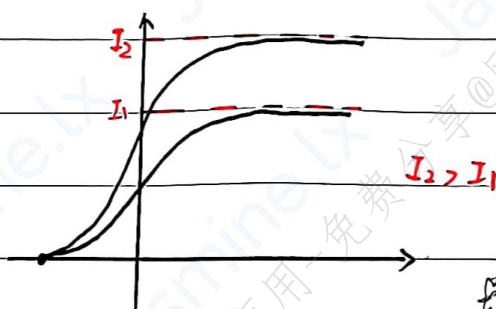
光电效应

当光照射到金属上时, 电子会从金属表面逸出的现象。

逸出的电子称为光电子。



a. 饱和光电流强度与入射光强成正比。



b. 对某种金属, 只有入射光频率大于

频率下限金属定

某一频率 ν_0 时, 电子才会从金属表面逸出, ν_0 称为截止频率或红限频率。

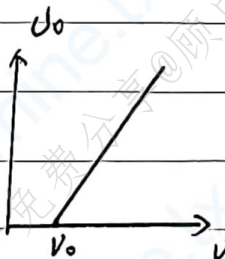
c. 遏制电势差

使光电流降为0所外加的反向电势差称为遏制电势差 U_0 。

$$\frac{1}{2} m v_m^2 = e U_0$$

遏制电势差与入射光频率具有线性关系

$$U_0 = k\nu - U_c$$



d. 瞬时性

1. 光子假说

光可视为由光子组成的粒子流, 单个光子的能量为 $\varepsilon = h\nu$

$$\text{红光: } \varepsilon_r = h\nu_r = h \frac{c}{\lambda_r} = 6.63 \times 10^{-34} \times \frac{3 \times 10^8}{760 \times 10^{-9}} \times \frac{1}{1.6 \times 10^{-19}} \\ = 1.64 \text{ eV}$$

$$\text{紫光: } \varepsilon_p = h\nu_p = 3.1 \text{ eV}$$

人眼可见光光子能量范围 (1.64 eV, 3.1 eV)

2. 爱因斯坦光电效应方程

$$h\nu = \frac{1}{2}mv^2 + W$$

→ 逸出功与材料有关

光 e^- 对逸出功 W

理论解释

a. 饱和电流

光强越大, 光子数越多, 单位时间内产生光电子数目越多, 光电流越大 ($\nu > \nu_0$)

$$I = n h \nu$$

b. 频率限制

$\nu > \nu_0$ 时才发生 $W = h\nu_0$ $\nu_0 = w/h$

c. 瞬时性 → (无积累)

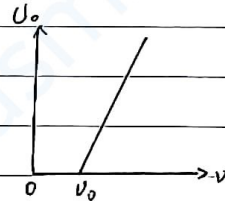
光子射到金属表面, 一个光子能量 $h\nu$

普朗克常数的测定

$$\begin{cases} h\nu = \frac{1}{2}mv^2 + W \\ \frac{1}{2}mv^2 = e \cdot U \end{cases}$$

$$\Rightarrow U = \frac{h}{e} \cdot \nu - \frac{W}{e} \quad k = \Delta U / \Delta \nu = h/e$$

$$\therefore h = ke$$



eg. 1



$$E_{\text{总}} = P \cdot t = P$$

$$\text{半径为 } r \text{ 的球面上 } E = \frac{P}{4\pi r^2} \cdot S$$

$$N = \frac{E}{h\nu} = \frac{E\lambda}{hc}$$

$$\sqrt{J} \cdot \frac{1}{e} \Rightarrow \text{eV}$$

eg. 2. 钾 红限波长 $\lambda_0 = 0.62 \mu\text{m}$

$$(1) \text{逸出功 } W = h\nu_0 = \frac{hc}{\lambda_0} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{0.62 \times 10^{-6}} \times \frac{1}{1.6 \times 10^{-19}} = 2 \text{ eV}$$

$$(2) \frac{1}{2}mv^2 = h\nu - h\nu_0 = h \cdot \frac{c}{\lambda} - h\nu_0 = 3.75 - 2 = 1.75 \text{ eV}$$

$$\begin{cases} \frac{1}{2}mv^2 = eU \\ \therefore U = \frac{\frac{1}{2}mv^2}{e} = \frac{1.75 \text{ eV}}{e} = 1.75 \text{ V} \end{cases}$$

e.g. 3.

光的波粒二象性

1. 波动性: 光的干涉和衍射

2. 粒子性: 光电效应

$$E = h\nu$$

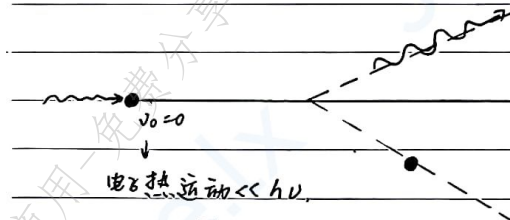
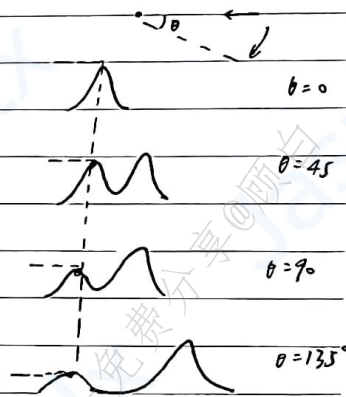
$$E = pc \quad p = E/c = h\nu/c = h/\lambda$$

$$\left. \begin{array}{l} E = h\nu \\ p = h/\lambda \end{array} \right\} \text{波动性}$$

§15-3 康普顿效应

波长偏移 $\Delta\lambda = \lambda - \lambda_0$ 与散射角无关

$\Delta\lambda$ 与散射物体无关



近认为静止电子

电子反冲速度很大, $\rightarrow$ 相对论力学

定量计算

能量守恒:

$$h\nu_0 + m_0c^2 = h\nu + mc^2$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

反冲电子所获动能

$$E_k = mc^2 - m_0c^2 = h\nu_0 - h\nu$$

动量守恒:

$$\frac{h\nu_0}{c} \frac{c}{\lambda_0} = \frac{h\nu}{c} \frac{c}{\lambda} + m\vec{v}$$

推导可得:

波长偏移量

$$\Delta\lambda = \frac{h}{m_0c} (1 - \cos\theta) = \lambda_c (1 - \cos\theta)$$

康普顿波长

$$\Delta\lambda = \lambda(\theta)$$

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① $\lambda = 400 \text{ nm}$ $\theta = \pi$ $\Delta\lambda = 2\lambda_c = 4.8 \times 10^{-3} \text{ nm}$

$\frac{\Delta\lambda}{\lambda_0} \sim 10^{-5}$ 偏移不明显

② X射线 $\lambda_0 = 0.04 \text{ nm}$ $\theta = \pi$ $\Delta\lambda = 4.8 \times 10^{-3} \text{ nm}$

$\frac{\Delta\lambda}{\lambda_0} \sim 10\%$ 偏移明显

$\lambda > \lambda_0$, $v < v_0$ 散射光能量减小.

a. 光具有波粒二象性. 光在传递过程中, 波动性较显著.

光与物质相互作用, 粒子性较显著.

b. $\lambda_0 \gg \lambda_c$, 则 $\lambda \approx \lambda_0$. 可见光观察不到康普顿效应.

c. $\Delta\lambda$ 与 θ 的关系与物质无关, 是光子与近自由电子间的相互作用.

d. 散射中 $\Delta\lambda = 0$ 的散射光是因光子与紧束缚电子的作用.

原子量大的物质, 其电子束缚较强, 因而康普顿效应不明显.

e.g., $\lambda_0 = 10^{-10} \text{ m}$ 与静止自由电子碰撞 $\theta = 90^\circ$

1. $\Delta\lambda = \lambda_c(1 - \cos\theta) = \lambda_c = 2.43 \times 10^{-12} \text{ m}$.

2. 反冲电子得到多少动能:

$E_k = mc^2 - m_0c^2 = \frac{hc}{\lambda_0} - \frac{hc}{\lambda} = \frac{hc}{\lambda_0} (1 - \frac{\lambda_0}{\lambda}) = 295 \text{ eV}$.

3. 在碰撞中, 光子的能量损失多少?

光子损失的能量 = 反冲电子的动能

§ 15-4 氢原子的玻尔理论

紫外光 莱曼系

$\sigma = \frac{1}{\lambda} = R(\frac{1}{1^2} - \frac{1}{n^2})$ $n = 2, 3, \dots$

可见光 巴耳末系

$\sigma = \frac{1}{\lambda} = R(\frac{1}{2^2} - \frac{1}{n^2})$ $n = 3, 4, \dots$

红外线 帕邢系

$\sigma = \frac{1}{\lambda} = R(\frac{1}{3^2} - \frac{1}{n^2})$ $n = 4, 5, \dots$

玻尔理论

a. 定态假设

电子在原子是沿轨道运动而不辐射电磁波

b. 量子化假设

$\checkmark \quad \angle = mvr = n\hbar = n \cdot \frac{h}{2\pi}$ $\hbar = 1.05 \times 10^{-34} \text{ J}\cdot\text{s}$

c. 频率假设

经典力学 $\frac{e^2}{4\pi\epsilon_0 r_n^2} = m \frac{v_n^2}{r_n}$

量子化条件 $m v_n r_n = n \hbar / 2\pi$

$\checkmark \quad r_n = \frac{\epsilon_0 \hbar^2}{\pi m e^2} n^2 = a_0 n^2$ $(n = 1, 2, \dots)$

$\checkmark$ 玻尔半径 $a_0 = 5.29 \times 10^{-11} \text{ m} = 0.529 \text{ \AA}$
 $\downarrow$
 10^{-10} m

能量

$E_n = \frac{1}{2} m v_n^2 - \frac{e^2}{4\pi\epsilon_0 r_n}$ $\checkmark \quad E_n = -\frac{me^4}{8\epsilon_0^2 \hbar^2} \cdot \frac{1}{n^2} = -\frac{E_1}{n^2}$

基态 ($n=1$) $E_1 = -me^4 / 8\epsilon_0^2 \hbar^2 = -13.6 \text{ eV}$ (电离能)

激发态 ($n>1$): $E_n = E_1 / n^2$

- 玻尔理论:
- 指出原子能级的存在 (原子能量量子化).
 - 指出定态和自动量子化的概念.
 - 正负角动量, 氢原子及类氢离子的光谱规律.

德布罗意公式:

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\nu = \frac{E}{h} = \frac{mc^2}{h}$$

德布罗意波或物质波.

§ 15-8

波函数

$$\psi(x, t) = \psi \cdot e^{-i \frac{\hbar}{\hbar} (Et - px)}$$

统计解释

体积元 dv 中粒子的概率 $|\psi|^2 dv = \psi \psi^* dv$

德布罗意波是概率波

上帝不在掷骰子.

归一化条件:

$$\int |\psi|^2 dv = 1 \quad (\text{束缚态})$$

标准条件:

波函数必须是单值, 连续, 有限的函数.

A 与 n 无关 $A = \sqrt{\frac{2}{a}}$

e.g. $\psi_n(x) = A \sin \frac{n\pi}{a} x \quad (0 \leq x \leq a) \quad \text{求 } n=1 \text{ 时}$

(1) $A =$

$\int |\psi|^2 dv = \int_0^a \psi(x) = A \sin \frac{\pi}{a} x$

$$\int_0^a |\psi|^2 dx = 1 \quad A^2 \int_0^a \sin^2 \frac{\pi}{a} x dx = \frac{a}{2} A^2 \int_0^a \sin^2 \frac{\pi}{a} x d \frac{\pi}{a} x$$

$$= \frac{a}{2} A^2 \int_0^{\pi} \sin^2 t dt$$

$$= \frac{a}{2} A^2 \cdot \frac{1}{2} \cdot \frac{\pi}{2} = 1$$

$$A^2 = \frac{2}{a} \quad A = \sqrt{\frac{2}{a}}$$

概率最大

$x = \frac{a}{2} \quad p(x) = \frac{2}{a}$

(2) $\int_0^a \frac{2}{a} \sin^2 \frac{\pi}{a} x dx = \frac{2}{a} \int_0^a \sin^2 \frac{\pi}{a} x d \frac{\pi}{a} x = \frac{2}{a} \int_0^{\pi} \sin^2 t dt = \frac{2}{a} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = 1$